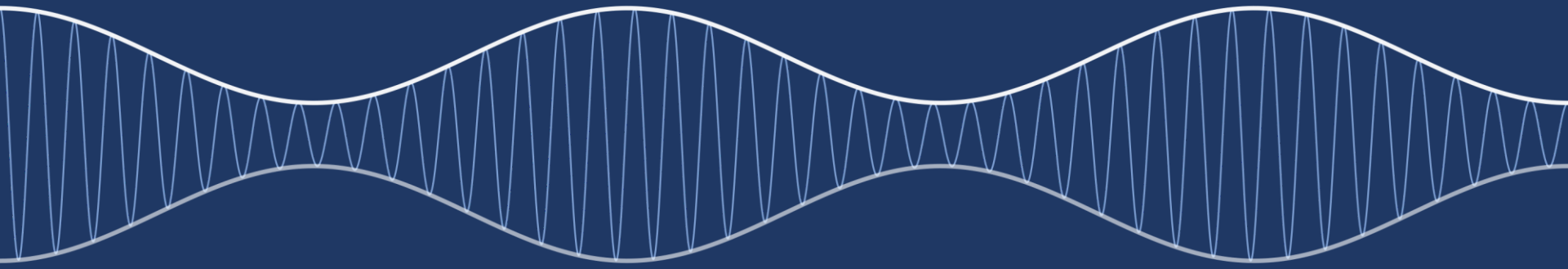
BRAC UNIVERSITY

Department of Electrical and Electronic Engineering, School of Engineering

Modulation and the Amplitude Modulation Family

EEE341 / ECE341

Introduction to Communication Engineering



Lecture Outline

1. **Modulation**
   Baseband vs. passband transmission, why we modulate, multiplexing, types of modulation, A/D conversion and PCM

2. **DSB with carrier (AM)**
   Time and frequency domain, modulation index, power efficiency, generation and detection

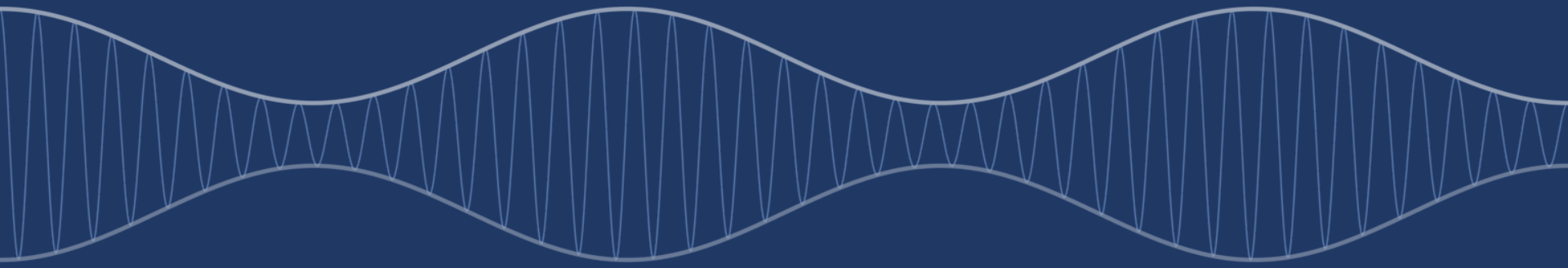
3. **DSB suppressed carrier**
   Principle, coherent detection, multiplier, nonlinear and switching modulators

4. **Bandwidth-efficient AM**
   SSB-SC, comparison with AM and DSB-SC, QAM, VSB and its TV applications

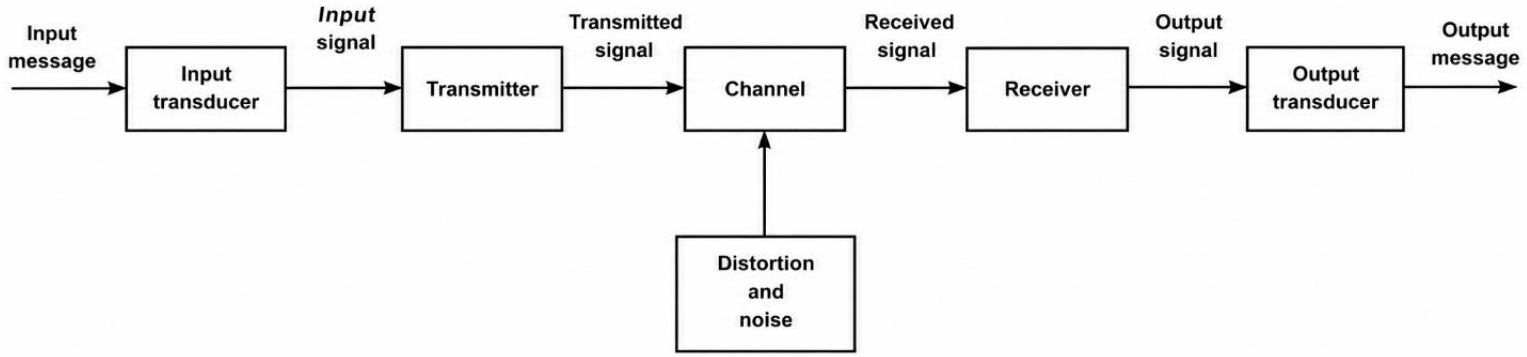
Worked examples: antenna size (Lecture Note), single-tone AM power, and VSB output-filter design.

Modulation

Why we modulate, and the different ways of doing it



The Communication System



Source

Originates the message (voice, picture, e-mail, data). A non-electric message is converted by an **input transducer** into the **baseband (message) signal**.

Transmitter

Modifies the baseband signal for efficient transmission: A/D converter, encoder and **modulator**.

Channel

Medium that carries the signal: twisted pair, coaxial cable, optical fiber or radio link. It **attenuates and distorts** the signal, and noise is added along the path.

Receiver

Reverses the transmitter's operations (**demodulator**, decoder, D/A converter) and removes channel distortion.

Destination

The unit to which the message is delivered, after an output transducer.

Modulation and Detection

Signals produced by message sources, or by A/D conversion, are called **baseband signals** because they are typically **low-pass** in nature.

Baseband signals may be transmitted directly over a suitable channel (e.g., telephone, fax). However, they are **not always suitable** for a given channel:

- When the signal and channel frequency bands do not match, **channels cannot be moved**.
- Hence, **messages must be moved** to the right channel frequency band.

In **modulation**, the baseband signal modifies some parameter of a high-frequency radio-frequency (RF) **carrier** (a sinusoid): its **amplitude, frequency or phase** is varied in proportion to m(t).

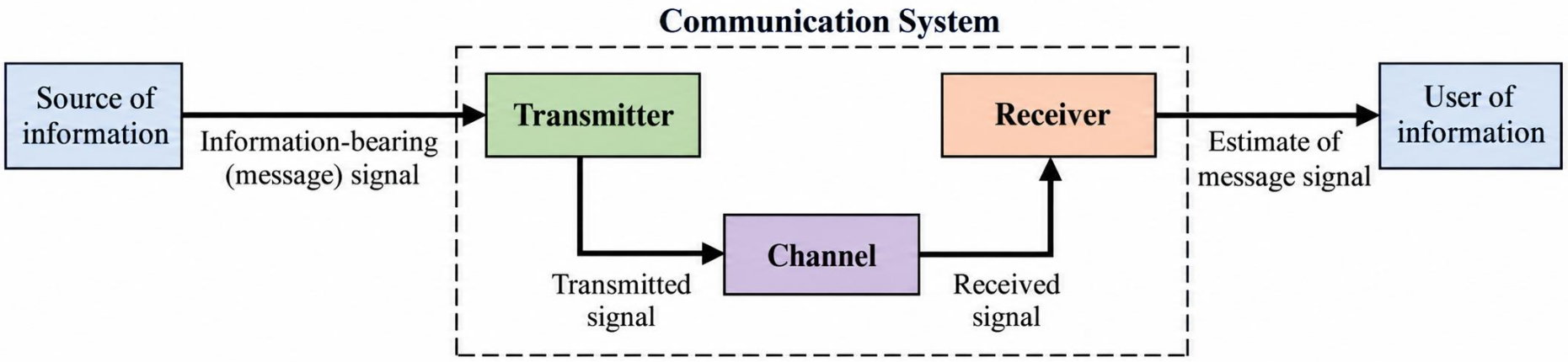
AM **Amplitude modulation**
carrier amplitude $\propto$ m(t)

FM **Frequency modulation**
carrier frequency $\propto$ m(t)

PM **Phase modulation**
carrier phase $\propto$ m(t)

Demodulation is the reverse process at the receiver that reconstructs the baseband signal.

Types of Transmission



1. Baseband transmission

The message is sent in its **original frequency band**, without shifting its spectrum.

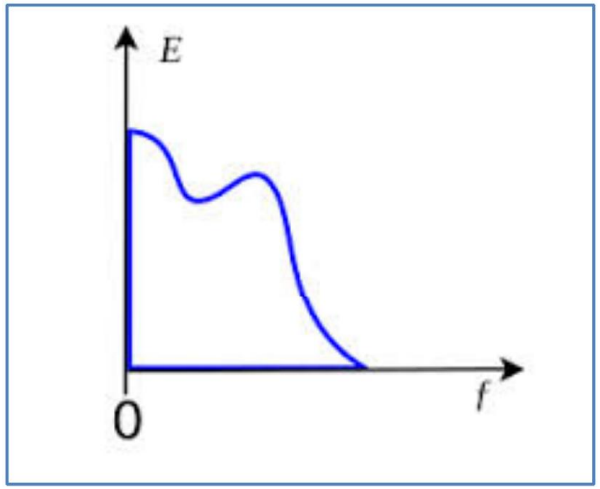
2. Passband transmission

Also called **carrier modulation**: the message is imposed on a high-frequency carrier, which shifts it to the channel's passband.

The rest of this lecture is about **passband transmission**, and in particular the amplitude modulation family.

Baseband Transmission

- **Baseband**: the frequency band of the original message signal from the source or input transducer.
- Most baseband signals (audio, video) contain a **significant low-frequency component**.
- A baseband bandwidth is equal to the **highest frequency** of a signal or system.
- **Baseband transmission** means transmitting the signal directly, **without any modification** to its spectral content.
- Baseband signals have **overlapping bands**, which results in severe interference if they share a channel:
  - cannot be effectively transmitted over the wireless channel
  - can be transmitted over copper or coaxial cable for a dedicated transmitter-receiver pair



Spectrum of a signal at baseband

Passband Transmission / Carrier Modulation

- The baseband signal is **imposed on a high-frequency carrier**.
- The amplitude, frequency or phase of the carrier is varied according to the amplitude of the baseband signal; this is called **modulation**.
- Modulation **shifts the baseband signal to a higher frequency range**.

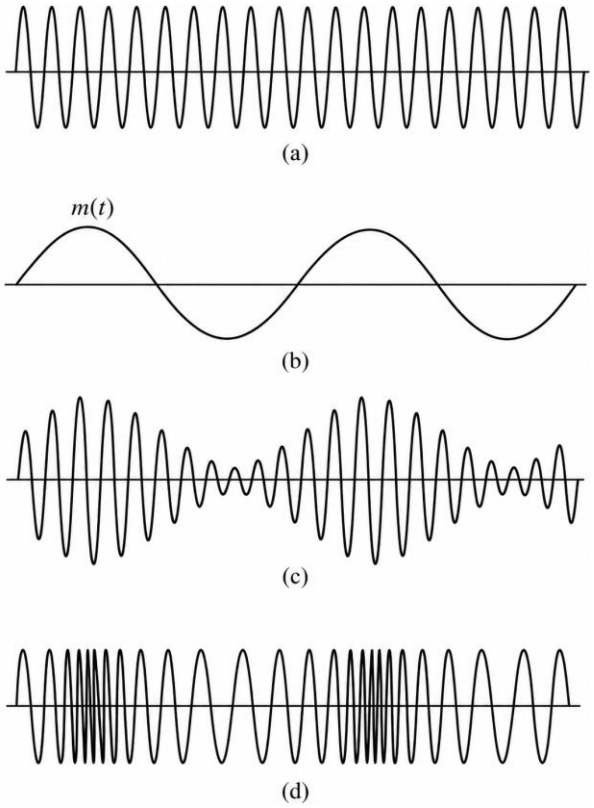
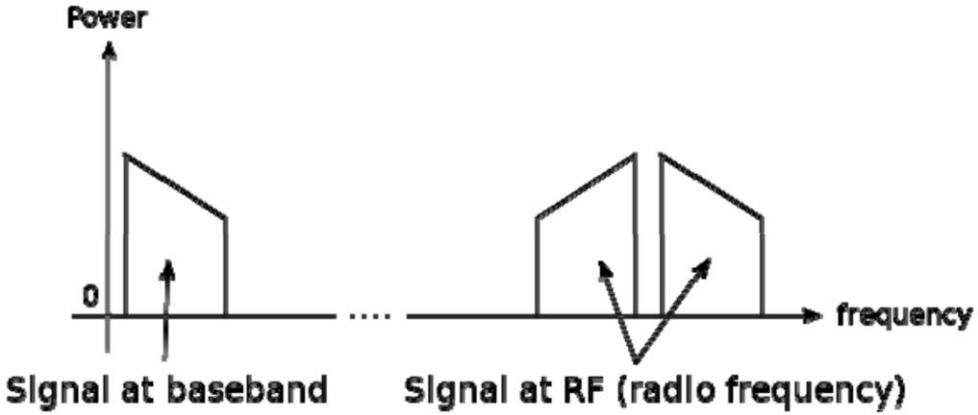
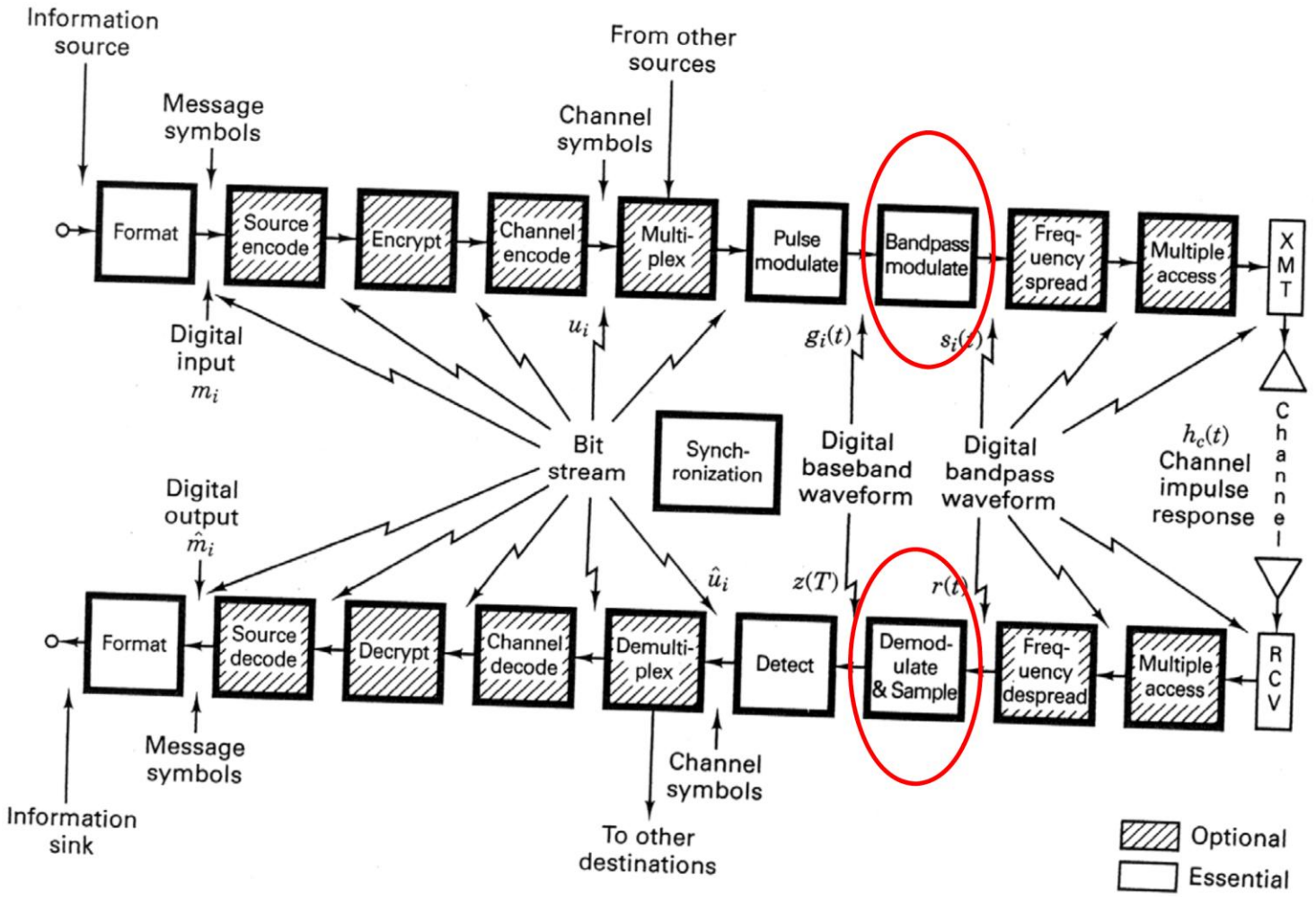


Figure 1.6. Modulation: (a) carrier; (b) modulating (baseband) signal; (c) amplitude-modulated wave; (d) frequency-modulated wave.



The same spectrum before and after modulation

A Comprehensive System Model



The bandpass modulator (transmitter) and the demodulator (receiver) are highlighted; hatched blocks are optional, white blocks essential.

Significance of Modulation (1/2)

Modulation is a process where some characteristic of a high-frequency carrier wave is varied in accordance with the amplitude of an information-bearing signal.

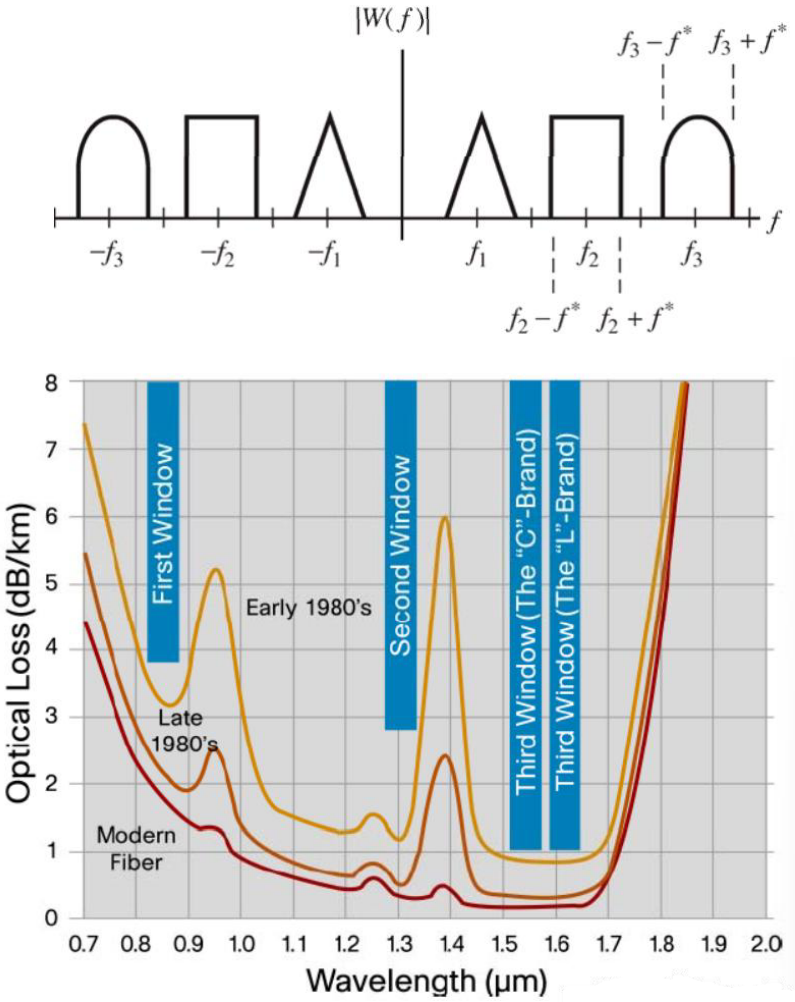
1. For efficient utilization of the frequency spectrum

- Use of the available frequency bands
- Simultaneous transmission of multiple users

2. To suit the channel requirement

A channel often acts like a **bandpass filter**, so the message spectrum must be shifted into the channel's passband. For instance:

- long-distance ionospheric propagation is possible for frequencies of 2 to 30 MHz; above 30 MHz, propagation is line of sight
- preferred frequencies for satellite communication are around 3 to 6 GHz
- optical fiber has low-loss windows only at certain wavelengths



Optical fiber attenuation windows

Multiplexing: Many Signals, One Channel

Modulation allows **multiple signals** to be transmitted at the same time in the same geographical area **without direct mutual interference**. Without it, all baseband video signals would interfere, because they effectively have the same bandwidth: cable or broadcast TV would be limited to one station at a time.

Frequency division multiplexing (FDM)

- Each TV station modulates a **different carrier frequency**, translating its signal to a different range.
- If carriers are far enough apart, the modulated spectra (TV channels) **do not overlap**.
- At the receiver, a **tunable bandpass filter** selects the desired channel.

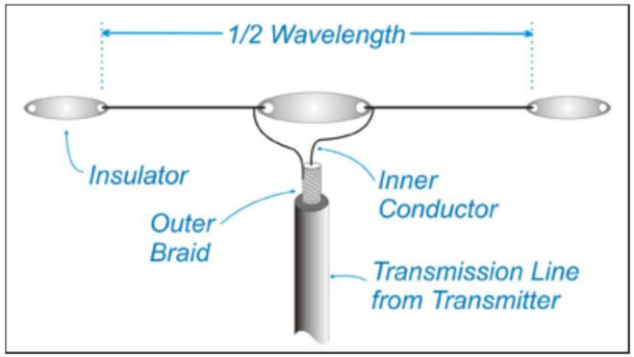
Time division multiplexing (TDM)

- Suitable when a signal is a **pulse train** (as in PCM).
- Narrower pulses leave gaps that are filled with **pulses from other signals**.
- The transmission time is shared by **interleaving** the pulse trains in a specified order; the receiver separates them.

Significance of Modulation (2/2)

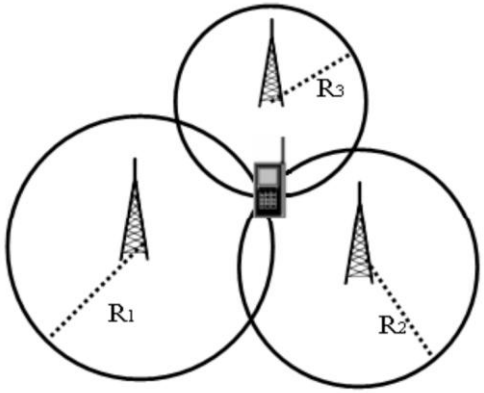
3. To decrease antenna length

- For transmitting a signal, the **antenna length must be of the order of a wavelength** (e.g., λ, λ/2 or λ/4).
- For a 1 Hz signal ($\lambda = 3 \times 10^8$ m), a λ/4 antenna would be **75,000 km** long!
- Modulated to 100 MHz (λ = 3 m), the required λ/4 antenna is only **0.75 m**.



4. To increase the operating range

- A signal requires **more energy to travel a longer distance**.
- Planck's formula: E = hv, where h is Planck's constant and v is the frequency of the signal.
- Modulation raises the frequency content of a signal, which indirectly increases its energy and lets it travel further.



$$P_r = P_t \left[\frac{\sqrt{G_l}\lambda}{4\pi d} \right]^2$$

Ease of Radiation: Why Baseband Needs a Carrier

- For efficient radiation of electromagnetic energy, the radiating antenna should be on the order of **a fraction or more of the wavelength** of the driving signal.
- For many baseband signals the wavelengths are **too large** for reasonable antenna dimensions.
- By modulating a high-frequency carrier, we **translate the signal spectrum** to the neighbourhood of the carrier frequency, which corresponds to a much smaller wavelength.
- In this respect, modulation is like letting the baseband signal **hitch a ride** on a high-frequency sinusoid (the carrier).

$$\lambda = \frac{c}{f}$$

$c = 3 \times 10^8$ m/s (speed of light)

A stone and a piece of paper

Thrown on its own, a piece of paper cannot go far. Wrapped around a stone (the carrier), it can be thrown much further.

Example: Antenna Size for a Speech Signal

PROBLEM (Lecture Note, Section 1.4.1)

The power in a speech signal is concentrated at frequencies from 100 Hz to 3000 Hz.

(a) Find the corresponding range of wavelengths.

(b) Comment on the antenna needed to radiate this signal directly.

(c) If the speech signal modulates a 10 MHz carrier, find the carrier wavelength and a practical antenna size.

Solution: Antenna Size for a Speech Signal

1

(a) Wavelength of the baseband speech signal, $\lambda = c / f$

f = 100 Hz: $\lambda = 3 \times 10^8 / 100 = 3 \times 10^6$ m = **3000 km**

f = 3000 Hz: $\lambda = 3 \times 10^8 / 3000 = 10^5$ m = **100 km**

2

(b) Antenna for direct (baseband) radiation

Even a quarter-wave antenna would be $\lambda/4$ = **25 km to 750 km** long: an **impractically large** antenna.

3

(c) After modulating a 10 MHz carrier

$\lambda_c = 3 \times 10^8 / (10 \times 10^6)$ = **30 m**

Antenna ≈ $\lambda_c/10$ = **3 m** (a quarter-wave antenna: $\lambda_c/4$ = 7.5 m)

Modulation shrinks the antenna from hundreds of kilometres to about 3 m

The spectrum is translated to around 10 MHz, so the antenna only has to suit the carrier's wavelength, not the speech signal's.

Demodulation

Once multiple modulated signals have arrived at the receiver, the desired signal must be **detected and recovered** into its original baseband form.

1. **Select**
   Because of FDM, the first stage of a demodulator is typically a **tunable bandpass filter** that selects the modulated signal in the band assigned to the station or channel.

2. **Convert**
   The demodulator then converts the carrier's variation of **amplitude, frequency or phase** back into the baseband signal voltage.

3. **Match the scheme**
   For AM, FM and PM, the detector output voltage must vary in proportion to the input signal's **amplitude, frequency and phase**, respectively.

4. **Recover**
   Such demodulators **down-convert** the RF signal back to the baseband signal that represents the original message: audio, video or data.

Types of Modulation

1. Analog / continuous-wave

Analog baseband signal on an analog carrier (bandpass channel)

A. Amplitude modulation (AM)

B. Angle modulation

- Frequency modulation (FM)
- Phase modulation (PM)

2. Digital modulation

Digital bit stream on an analog carrier (bandpass channel)

A. Amplitude shift keying (ASK)

B. Frequency shift keying (FSK)

C. Phase shift keying (PSK)

3. Pulse modulation

Analog narrowband signal on pulses (wideband baseband channel)

A. Pulse amplitude modulation (PAM)

B. Pulse time modulation (PTM)

- Pulse width / duration (PWM / PDM)
- Pulse position (PPM)

C. Pulse code modulation (PCM)

D. Delta modulation (DM)

Pulse modulation: **A, B** are analog pulse modulation; **C, D** are digital pulse modulation.

Analog-to-Digital (A/D) Conversion

Analog signals are continuous in time and in range; digital signals exist **only at discrete times** and take **only finite values**. A/D conversion is the meeting ground between them.

1. Sampling

Sampling theorem: if the highest frequency in the signal spectrum is B Hz, the signal can be reconstructed from samples taken uniformly at a rate **not less than 2B samples per second**.

2. Quantization

If m(t) lies in ($-m_p$, m_p), the range is divided into **L intervals**; each sample is rounded to the **midpoint** of its interval, so it is one of L numbers.

A/D conversion is never 100% accurate, but human perception does not need infinite accuracy.

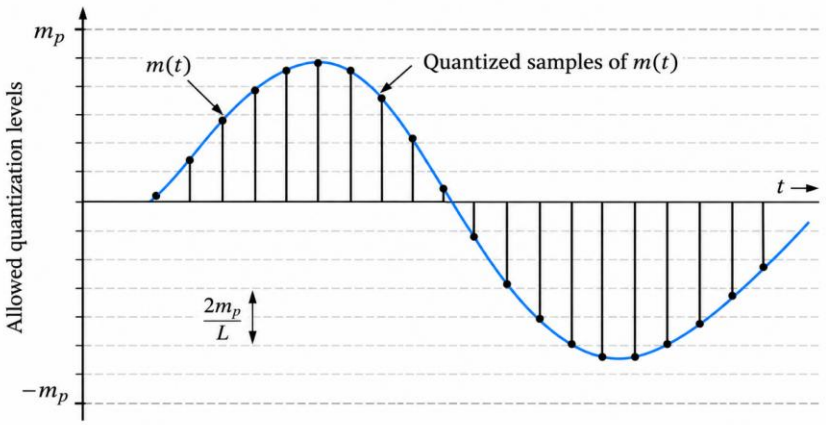


Figure 1.4. Analog-to-digital conversion of a signal

The price of digital communication: **increased processing complexity** and **transmission bandwidth**.

Pulse-Code Modulation (PCM)

After A/D conversion, each sample takes one of **L** quantization levels. PCM represents each quantized sample by an ordered combination of **two basic pulses**:

- $p_1(t)$ representing 1, and $p_0(t)$ representing 0

Each of the L levels can be written as a bit string of length **$\log_2 L$**, so each sample maps to a short sequence of pulses.

Signalling uses only two symbols. The **binary** case is of great practical importance because of its simplicity and ease of detection.

EXAMPLE (Lecture Note, Section 1.2.4)

L = 16 levels $\Rightarrow$ $\log_2 16$ = **4 bits** per sample

With $p_1(t) = A/2$ and $p_0(t) = -A/2$, four pulses give $2 \times 2 \times 2 \times 2$ = **16 distinct patterns**, one for each level (Fig. 1.5).

Digit	Binary equivalent	Pulse code waveform
0	0000	
1	0001	
2	0010	
3	0011	
4	0100	
5	0101	
6	0110	
7	0111	
8	1000	
9	1001	
10	1010	
11	1011	
12	1100	
13	1101	
14	1110	
15	1111	

Figure 1.5. Example of PCM encoding

The Amplitude Modulation (AM) Family

The **amplitude of the carrier** is varied in accordance with the instantaneous amplitude of the modulating (message) signal.

Double-sideband (DSB)

- DSB with carrier (DSB-WC) / AM
- DSB suppressed carrier (DSB-SC)
- DSB reduced carrier (DSB-RC)

Single-sideband (SSB)

- SSB with carrier (SSB-WC)
- SSB suppressed carrier (SSB-SC)

Vestigial sideband (VSB)

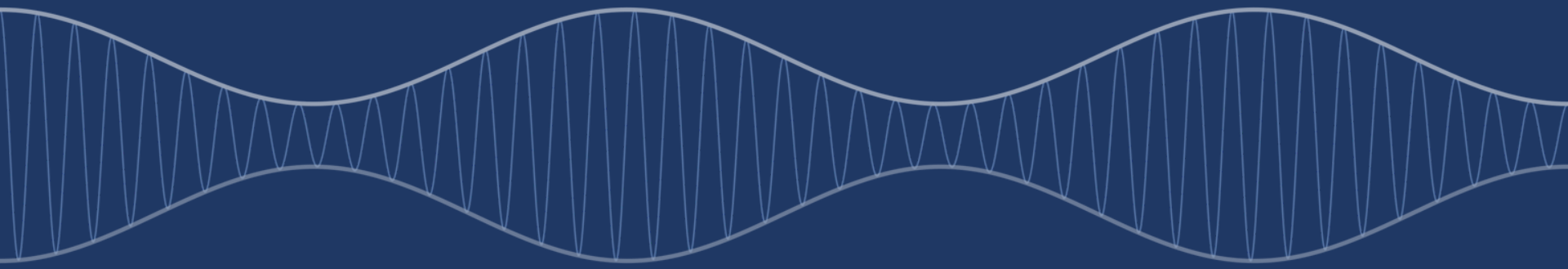
- A compromise between DSB and SSB

Quadrature AM (QAM)

- Two DSB-SC signals on carriers in quadrature

DSB with Carrier (DSB-WC / AM)

Time and frequency domain, modulation index, power efficiency, generation and detection

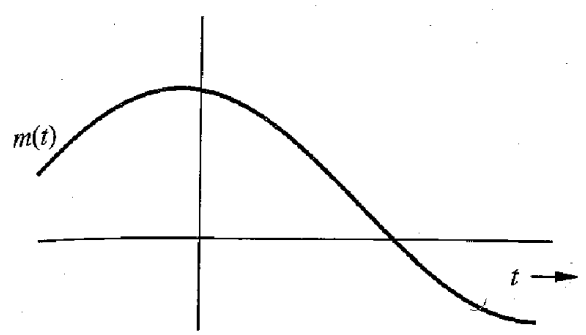


DSB-WC / AM (1): Time Domain

The carrier signal

$$c(t) = A_c \cos \omega_c t = A_c \cos 2\pi f_c t$$

f_c = carrier frequency (Hz)



The baseband message (modulating) signal: m(t)

The AM signal

modulated carrier

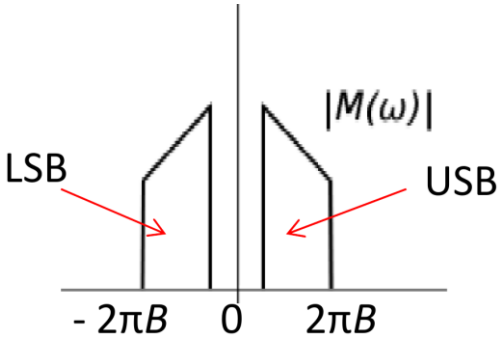
$$\varphi_{AM}(t) = [A_c + m(t)]\cos \omega_c t = A_c \cos \omega_c t + m(t)\cos \omega_c t$$

carrier

$$\varphi_{AM}(t) = A_c \left[1 + \frac{m(t)}{A_c}\right] \cos \omega_c t = A_c \, [1 + k_a m(t)] \cos \omega_c t$$

We transmit an **unmodulated carrier** in addition to the modulated carrier. $k_a = 1/A_c$: amplitude sensitivity of the modulator.

AM (2): Frequency Domain



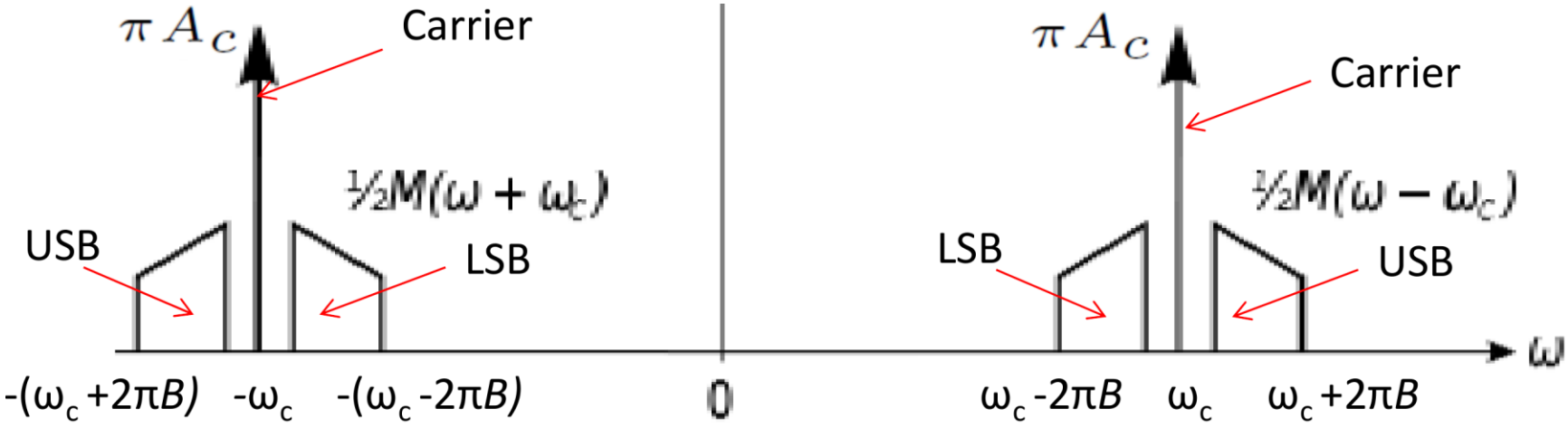
Message spectrum, bandwidth B Hz

Amplitude spectrum of the AM signal

$$\Phi_{AM}(f) = \frac{A_c}{2}[\delta(f-f_c)+\delta(f+f_c)] + \frac{1}{2}[M(f-f_c)+M(f+f_c)]$$

or, in terms of ω,

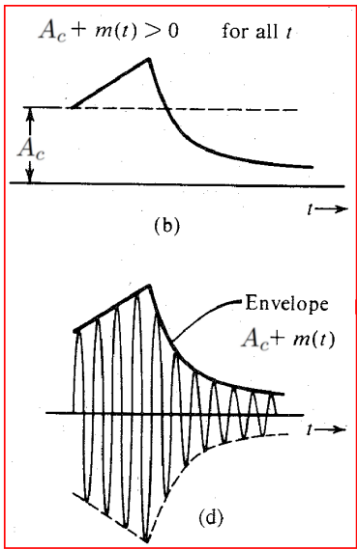
$$\Phi_{AM}(\omega) = \pi A_c[\delta(\omega-\omega_c)+\delta(\omega+\omega_c)] + \frac{1}{2}[M(\omega-\omega_c)+M(\omega+\omega_c)]$$

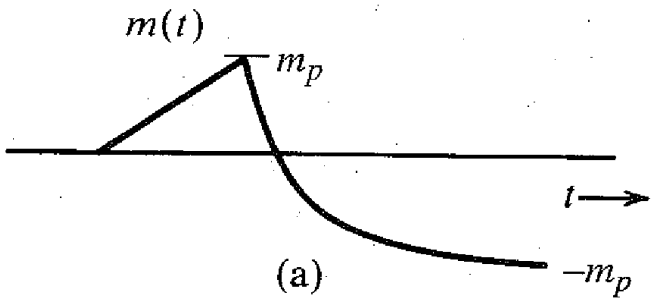


Message bandwidth B Hz ⇒ AM bandwidth B_{AM} = 2B Hz

AM (3): The Envelope

- Envelope of the modulated carrier: **$e(t) = A_c + m(t)$**
- The carrier oscillates between the envelope $|A_c + m(t)|$ and its negative image $-|A_c + m(t)|$.

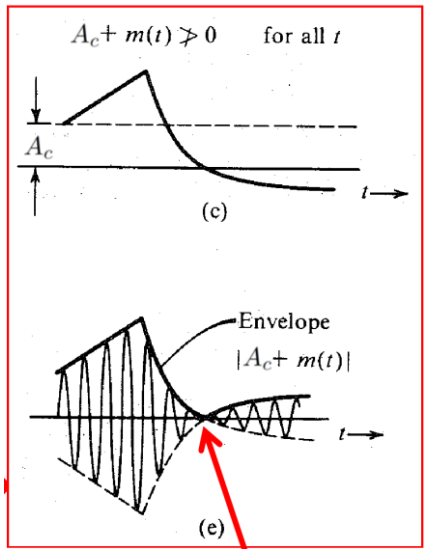




← Envelope detection can be used

Envelope detection cannot be used →

(note the 180° phase reversal)



The envelope is an accurate representation of the message, provided

(a) $f_c \gg B$ (B is the message bandwidth): avoids overlap of the frequency spectrum components

(b) $A_c + m(t) \geq 0$ for all t: ensures the message can be recovered from the envelope

AM (4): Modulation Index, μ

Definition

$$\mu = \frac{e_{max} - e_{min}}{e_{max} + e_{min}}$$

Max. and min. of the envelope:

$e_{max} = A_c + m(t)|_{max}$

$e_{min} = A_c + m(t)|_{min}$

Case I: m(t) with zero offset

$m(t)|_{max} = m_p = -m(t)|_{min}$

$$\mu = \frac{m_p}{A_c}$$

Case II: non-zero offset (rare)

$m(t)|_{max} \neq -m(t)|_{min}$

$$\mu = \frac{m(t)|_{max} - m(t)|_{min}}{2A_c + m(t)|_{max} + m(t)|_{min}}$$

For envelope detection: $A_c + m(t)|_{min} \geq 0$

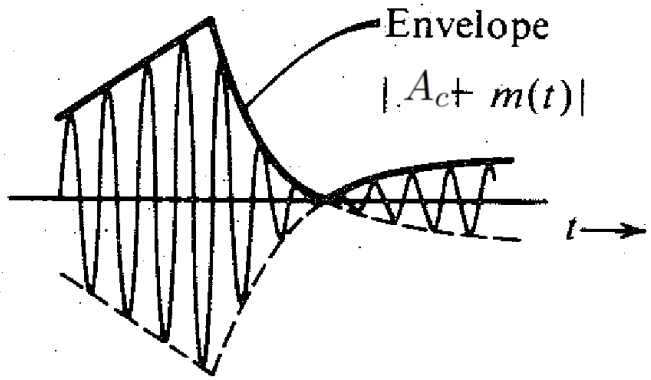
Case I: $A_c - m_p \geq 0 \quad \Rightarrow \quad 0 \leq \mu \leq 1$

Case II: solve from the same envelope-detection condition (exercise).

μ < 1 — **Undermodulation**
envelope detection possible

μ = 1 — **100% modulation**
envelope detection possible

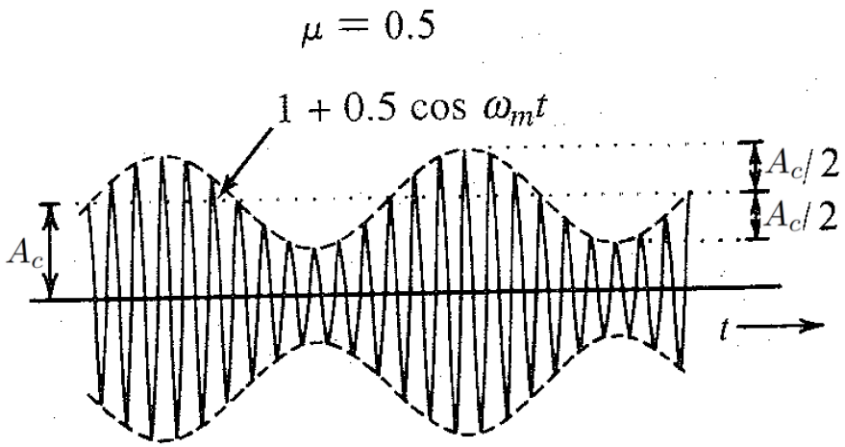
μ > 1 — **Overmodulation**
envelope detection creates distortion



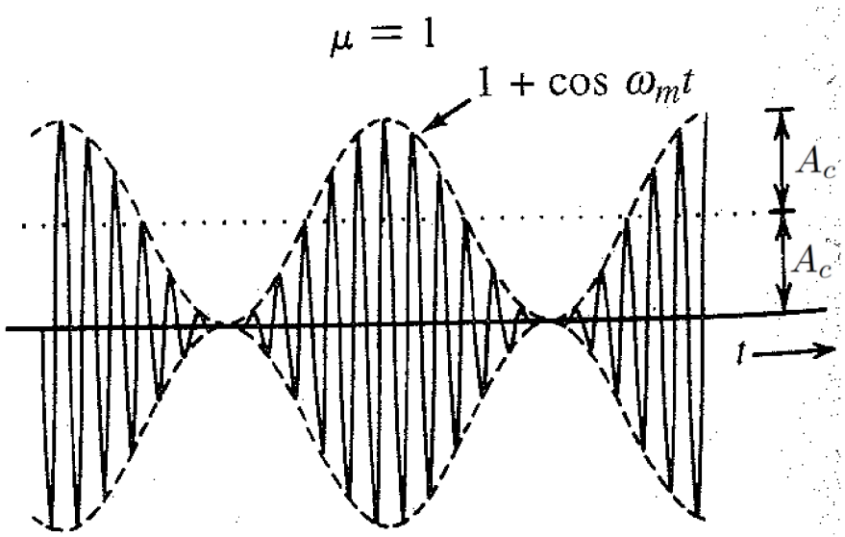
An example of overmodulation

AM (5): Single-Tone Modulation

$$m(t) = A_m \cos \omega_m t \quad \Rightarrow \quad \varphi_{AM}(t) = A_c\,[1 + \mu \cos \omega_m t] \cos \omega_c t, \qquad \mu = \frac{A_m}{A_c}$$



μ = 0.5: 50% modulation



μ = 1: 100% modulation

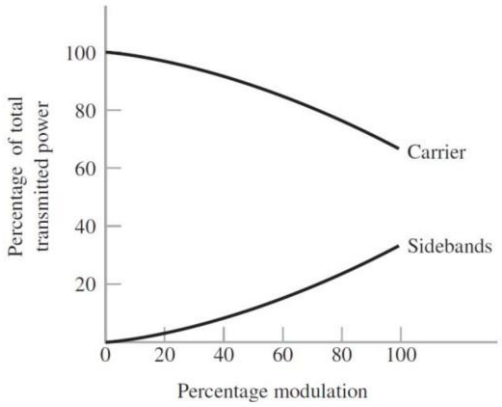
Try it: sketch the amplitude spectrum of $\phi_{AM}(t)$. Expand the product.

AM (6): Carrier and Sideband Power

Transmitted signal $\varphi_{AM}(t) = A_c \cos \omega_c t + m(t) \cos \omega_c t$

Carrier and sideband power $P_c = \frac{A_c^2}{2}$ $\quad P_s = \frac{1}{2}\overline{m^2(t)}$

Power (modulation) efficiency $\eta = \frac{\text{useful power}}{\text{total power}} = \frac{P_s}{P_c + P_s}$



Share of total power vs. % modulation

Special case: single-tone modulation, $m(t) = A_m \cos \omega_m t$

$$\varphi_{AM}(t) = A_c \cos \omega_c t + \frac{\mu A_c}{2}[\cos(\omega_c + \omega_m)t + \cos(\omega_c - \omega_m)t]$$

$$P_{USB} = P_{LSB} = \frac{A_c^2 \mu^2}{8} \qquad P_s = \frac{A_c^2 \mu^2}{4} \qquad \eta = \frac{\mu^2}{2 + \mu^2} \times 100\%$$

η rises monotonically from 0 to 1/3 as μ goes from 0 to 1.

Thus, for tone modulation, under the best conditions (for envelope detection), only one-third of the power is used for carrying message, which is even lower (less than 25% or worse) under practical conditions

Additional Example: Single-Tone AM Power

PROBLEM

An AM transmitter has a carrier power of 10 kW and is modulated by a single tone with $\mu = 0.6$. Find the power in each sideband, the total transmitted power and the power efficiency.

1 **Sideband power ($P_s = \mu^2 A_c^2/4 = (\mu^2/2) P_c$)**

$P_s = (0.6^2 / 2) \times 10\ \text{kW} = 0.18 \times 10\ \text{kW} =$ **1.8 kW**, so $P_{USB} = P_{LSB} =$ **0.9 kW**

2 **Total transmitted power**

$P_T = P_c + P_s = 10 + 1.8 =$ **11.8 kW** (check: $P_T = (1 + \mu^2/2) P_c = 1.18 \times 10\ \text{kW}$)

3 **Power efficiency**

$\eta = P_s / P_T = 1.8 / 11.8 =$ **15.3%** (check: $\eta = \mu^2/(2 + \mu^2) = 0.36 / 2.36$)

Only about 15% of the power carries the message; 85% goes to the carrier

AM (7): Modulation by Multi-Tone Signals

Message signal

$$m(t) = A_1 \cos \omega_1 t + A_2 \cos \omega_2 t + A_3 \cos \omega_3 t + \cdots + A_n \cos \omega_n t$$

AM signal

$$\varphi_{AM}(t) = A_c \left[1 + \mu_1 \cos \omega_1 t + \mu_2 \cos \omega_2 t + \cdots + \mu_n \cos \omega_n t\right] \cos \omega_c t$$

Total sideband power

$$P_s = \left(\mu_1^2 + \mu_2^2 + \cdots + \mu_n^2\right) \frac{A_c^{\,2}}{4} = \mu_T^2 \frac{A_c^{\,2}}{4}$$

Total transmitted power

$$P_T = P_c + P_s = \left(1 + \frac{\mu_T^2}{2}\right) P_c \qquad \mu_T^2 = \mu_1^2 + \mu_2^2 + \mu_3^2 + \ldots + \mu_n^2$$

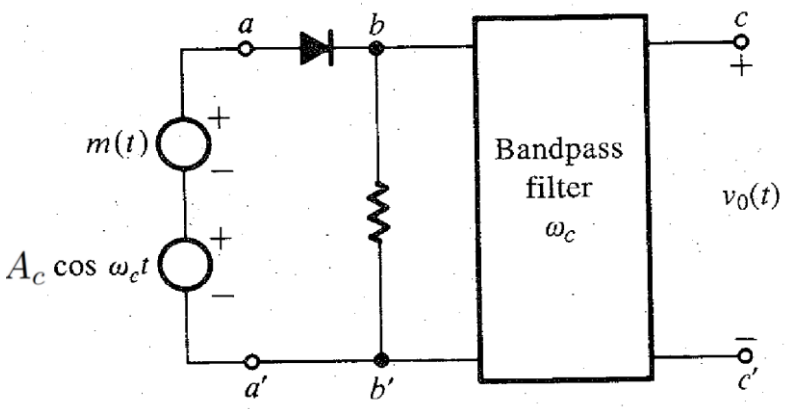
Overall (equivalent) modulation index

$$\mu_T = \sqrt{\mu_1^2 + \mu_2^2 + \mu_3^2 + \cdots + \mu_n^2}$$

Note 1: $\mu = (e_{max} - e_{min})/(e_{max} + e_{min})$ and μ_T may **not be equal**.

Note 2: $\mu_T \leq 1$ does **not** guarantee the prevention of overmodulation (it does not ensure envelope detection).

Generation of AM: Switching Modulator



Input: $A_c \cos \omega_c t + m(t)$

Condition: $A_c \gg m(t)$, so the diode's switching action is controlled by $A_c \cos \omega_c t$.

The switching multiplies $[A_c \cos \omega_c t + m(t)]$ by the **periodic pulse train w(t)**.

$$w(t) = \frac{1}{2} + \frac{2}{\pi}\left(\cos \omega_c t - \frac{1}{3}\cos 3\omega_c t + \frac{1}{5}\cos 5\omega_c t - \cdots\right)$$

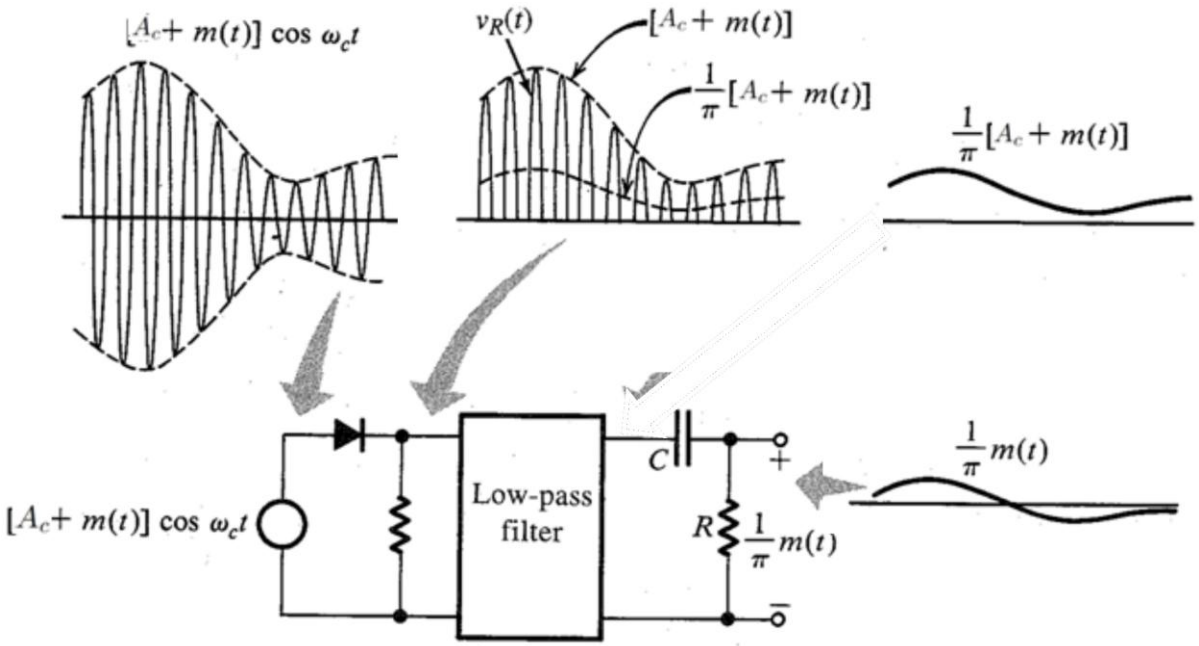


Thus, $$v_{bb'}(t) = [A_c \cos \omega_c t + m(t)]\, w(t)$$

and, after the bandpass filter, $$v_{cc'}(t) = \frac{A_c}{2}\cos \omega_c t + \frac{2}{\pi} m(t) \cos \omega_c t$$ **= an AM signal**

Demodulation of AM (1): Rectifier Detector

Valid for μ ≤ 1



$$v_R = \{[A_c+ m(t)] \cos \omega_c t\}\, w(t)$$

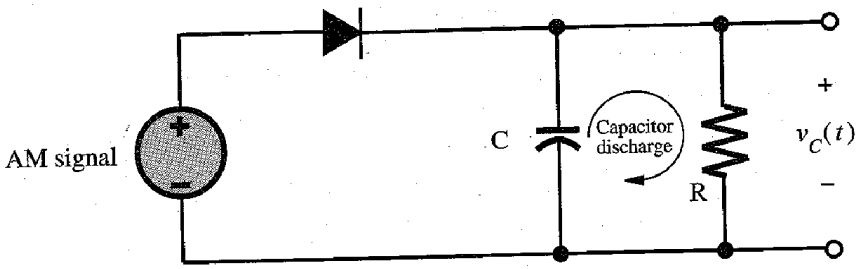
$$= [A_c+ m(t)] \cos \omega_c t \left[\frac{1}{2} + \frac{2}{\pi}\left(\cos \omega_c t - \frac{1}{3}\cos 3\omega_c t + \frac{1}{5}\cos 5\omega_c t - \cdots\right)\right]$$

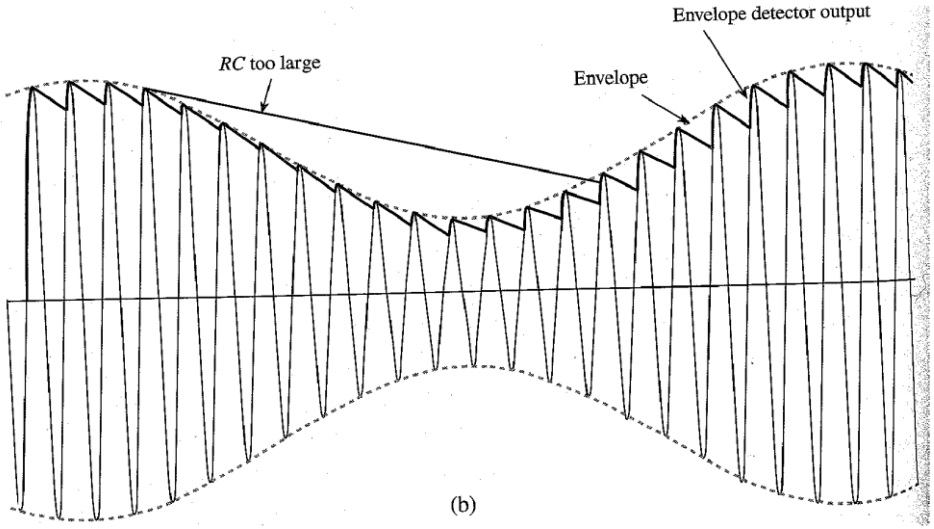
$$= \frac{1}{\pi}[A_c+ m(t)] + \text{other terms of higher frequencies}$$

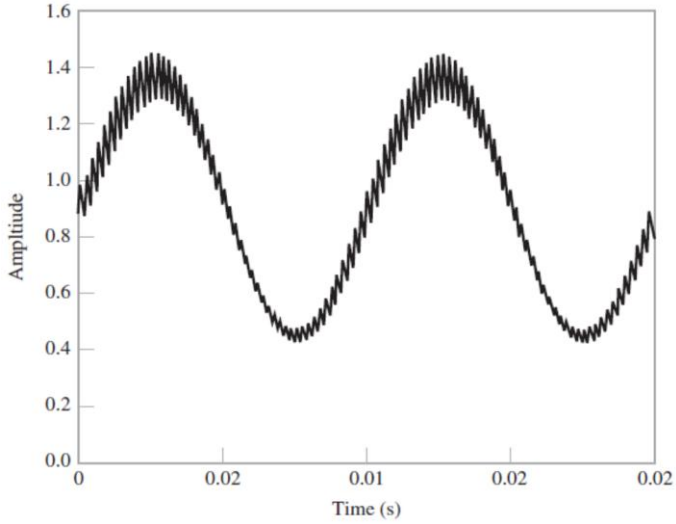
The low-pass filter removes the higher-frequency terms; the capacitor blocks the dc term A_c/π, leaving **m(t)/π**.

Demodulation of AM (2): Envelope Detector

Valid for μ ≤ 1







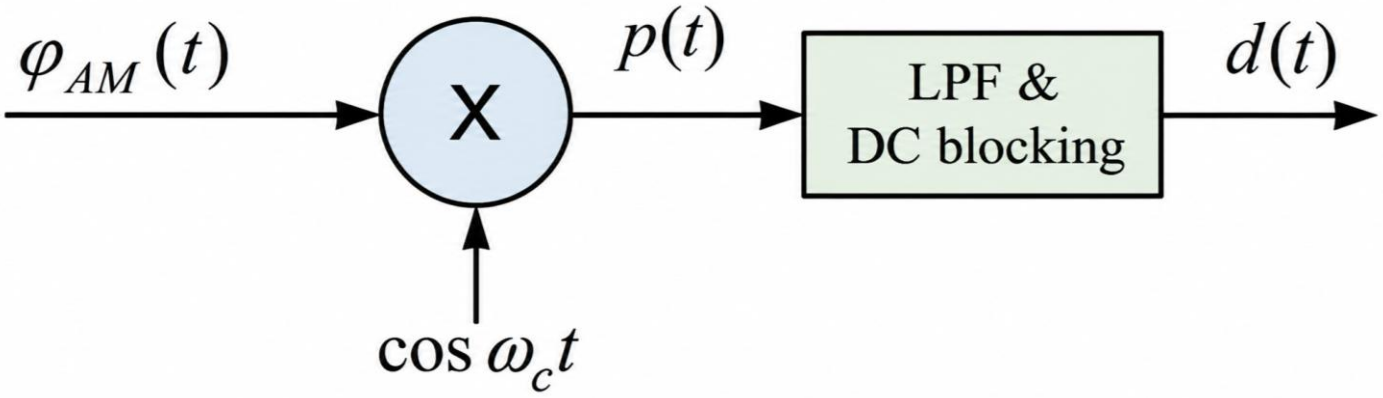
The detector output is $v_c(t) = A_c + m(t)$ with a **ripple at ω_c**. The dc term A_c is blocked by a capacitor or an RC high-pass filter; the ripple can be reduced by another low-pass filter.

Choose the discharge time constant RC properly:

$$\frac{1}{\omega_c} \ll RC < \frac{1}{2\pi B}$$

Demodulation of AM (3): Coherent Detector

Synchronous / coherent / homodyne detector: **no restriction on μ**



$$p(t) = \varphi_{AM}(t)\cos\omega_c t = [A_c + m(t)]\cos^2 \omega_c t$$

$$= \frac{1}{2}[A_c + m(t)](1 + \cos 2\omega_c t)$$

$$= \frac{A_c}{2} + \frac{1}{2}m(t) + \frac{1}{2}[A_c + m(t)]\cos 2\omega_c t$$

After the LPF and DC blocking:

d(t) = C m(t), where C is a constant.

- The phase and frequency of the local carrier must equal those of the carrier:
  - **synchronization** is required between transmitter and receiver.
- **More complex and expensive** than an envelope detector.
- The rectifier detector is effectively a coherent detector.

AM: Summary

Drawbacks

- **Wasteful of transmitted power**: power efficiency is very low (at most 33%).
- **Wasteful of channel bandwidth**: twice the message bandwidth.
- **Easily affected by noise**, since noise affects the amplitude easily.

Advantages

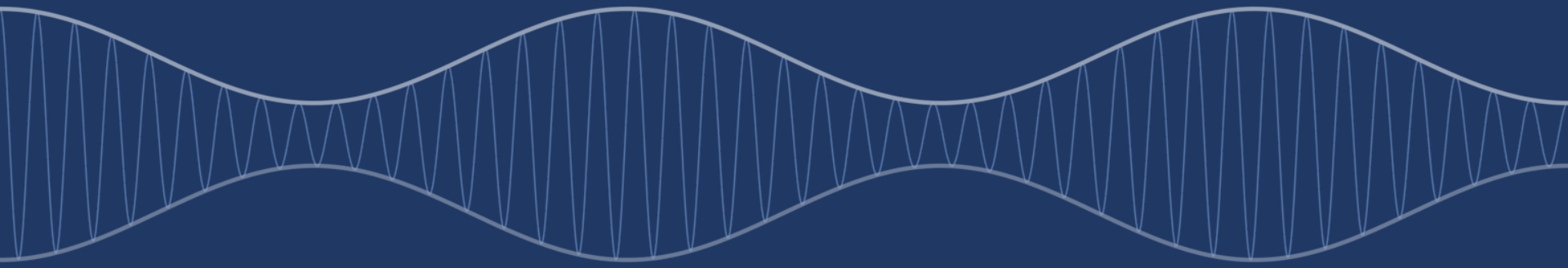
- **Simpler** modulator and demodulator can be used.
- **Less expensive** modulator and demodulator can be used.

Next: remove the carrier to save power (**DSB-SC**), then remove a sideband to save bandwidth (**SSB**).

PART 3

DSB Suppressed Carrier (DSB-SC)

Principle, coherent detection, and multiplier, nonlinear and switching modulators



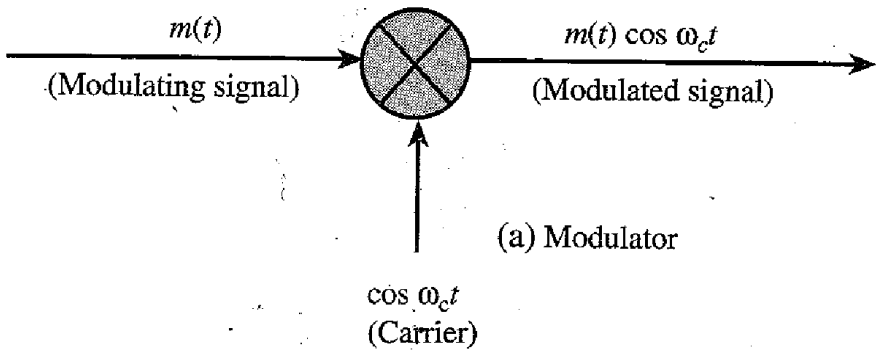
DSB-SC: Principle (1)

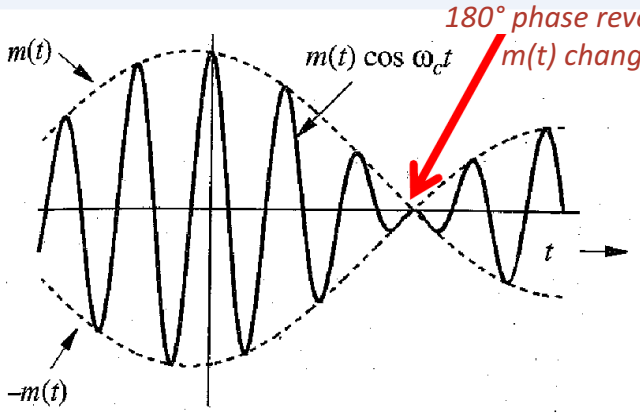
- The carrier is **suppressed** before transmission (not transmitted).
- **Power efficiency increases to 100%**: the transmit power requirement decreases.
- **Bandwidth requirement**: twice the bandwidth of the message signal.
- **Envelope detection cannot be used.**

DSB-SC signal

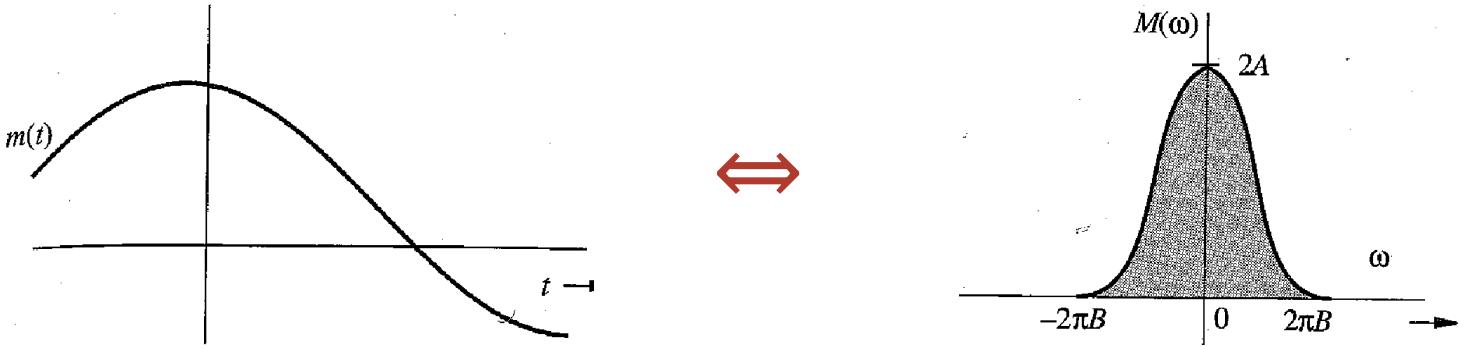
$$\varphi_{\mathrm{DSB-SC}}(t) = m(t)\cos\omega_c t \qquad \text{or} \qquad \varphi_{\mathrm{DSB-SC}}(t) = A_c\, m(t)\cos\omega_c t$$

DSB-SC signal generation



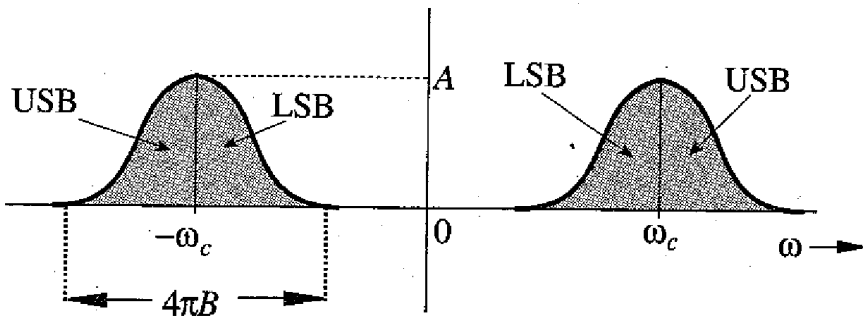


DSB-SC: Principle (2): Spectrum



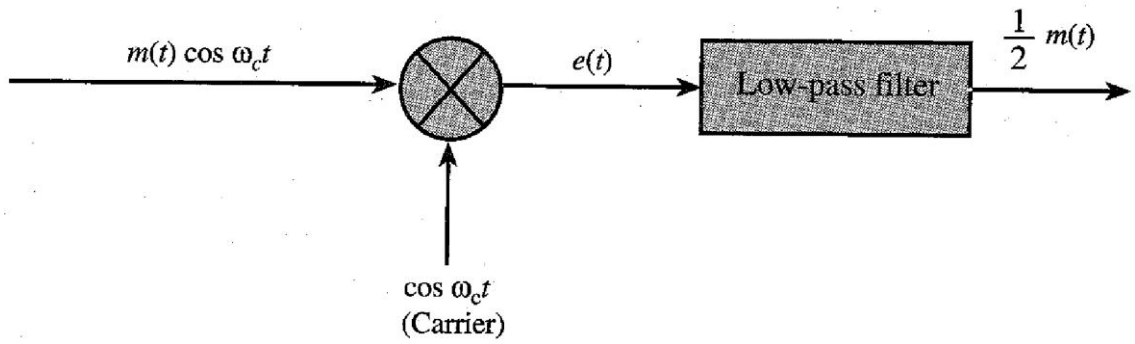
m(t) ⇔ M(ω)

$$m(t)\cos\omega_c t \iff \frac{1}{2}[M(\omega+\omega_c)+M(\omega-\omega_c)]$$



No carrier impulse: only the upper and lower sidebands, each copy 4πB wide (bandwidth 2B Hz)

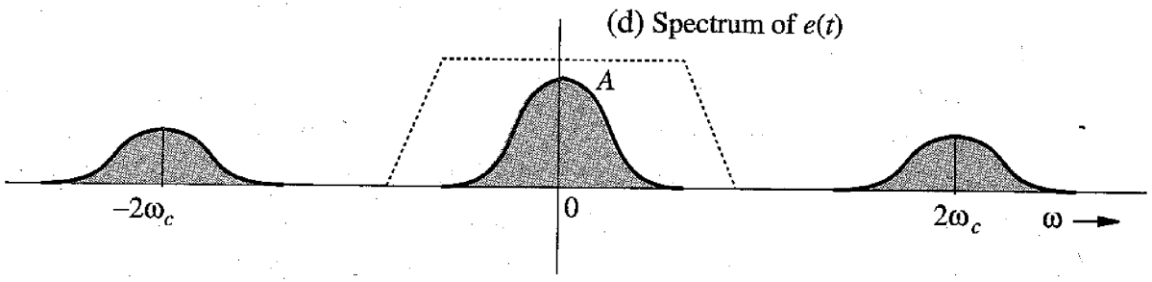
DSB-SC: Principle (3): Coherent Demodulation



Synchronous / coherent / homodyne demodulator (detector)

$$e(t) = \varphi_{\text{DSB-SC}}(t)\cos\omega_c t = m(t)\cos^2\omega_c t = \frac{1}{2}m(t) + \frac{1}{2}m(t)\cos 2\omega_c t$$

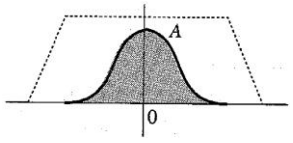
$$E(\omega) = \frac{1}{2}M(\omega) + \frac{1}{4}[M(\omega + 2\omega_c) + M(\omega - 2\omega_c)]$$



Spectrum of e(t), with the low-pass filter response dotted

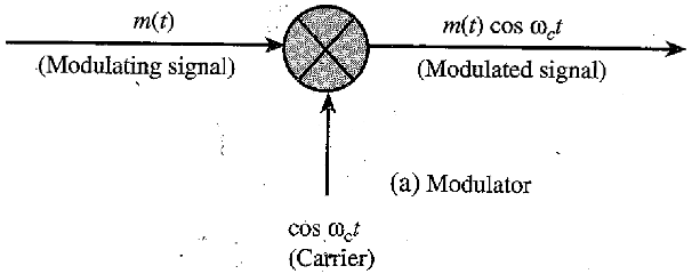
After the LPF:

(1/2) m(t)



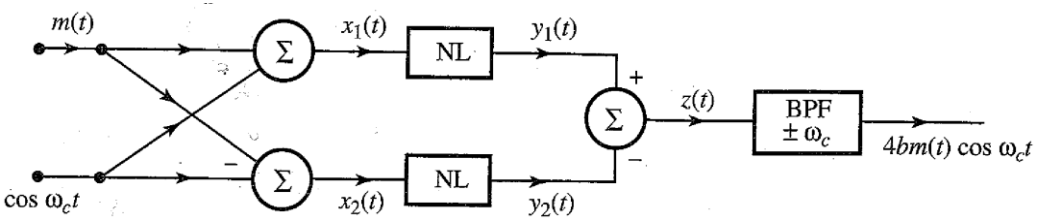
DSB-SC Modulators (1): Multiplier and Nonlinear

1. Multiplier modulator



Multiplies m(t) directly by the carrier. Simple in principle, but an accurate analog multiplier is hard to build at RF.

2. Nonlinear modulator



Nonlinear element characteristic:

$$y(t) = a\,x(t) + b\,x^2(t), \qquad x_1(t) = \cos\omega_c t + m(t), \qquad x_2(t) = \cos\omega_c t - m(t)$$

$$z(t) = y_1(t) - y_2(t) = 2a\,m(t) + 4b\,m(t)\cos\omega_c t$$

- The BPF removes 2a m(t) and passes **4b m(t) cos ω_ct**.
- The carrier does **not** appear at the input of the final BPF.
- The circuit is balanced for the carrier only, so it is a **single balanced modulator**.

DSB-SC Modulators (2): Switching Principle

3. Switching modulators

- The multiplication operation is replaced by a **simpler switching operation**.
- Multiplication can be achieved by multiplying m(t) by **any periodic signal ϕ(t)** of fundamental frequency ω_c:

$$\phi(t) = \sum_{n=0}^{\infty} C_n \cos(n\omega_c t + \theta_n)$$

$$m(t)\,\phi(t) = \sum_{n=0}^{\infty} C_n\, m(t) \cos(n\omega_c t + \theta_n)$$

The spectrum of m(t)ϕ(t) is M(ω) shifted to **$\pm\omega_c$, $\pm 2\omega_c$, …, $\pm n\omega_c$, …**

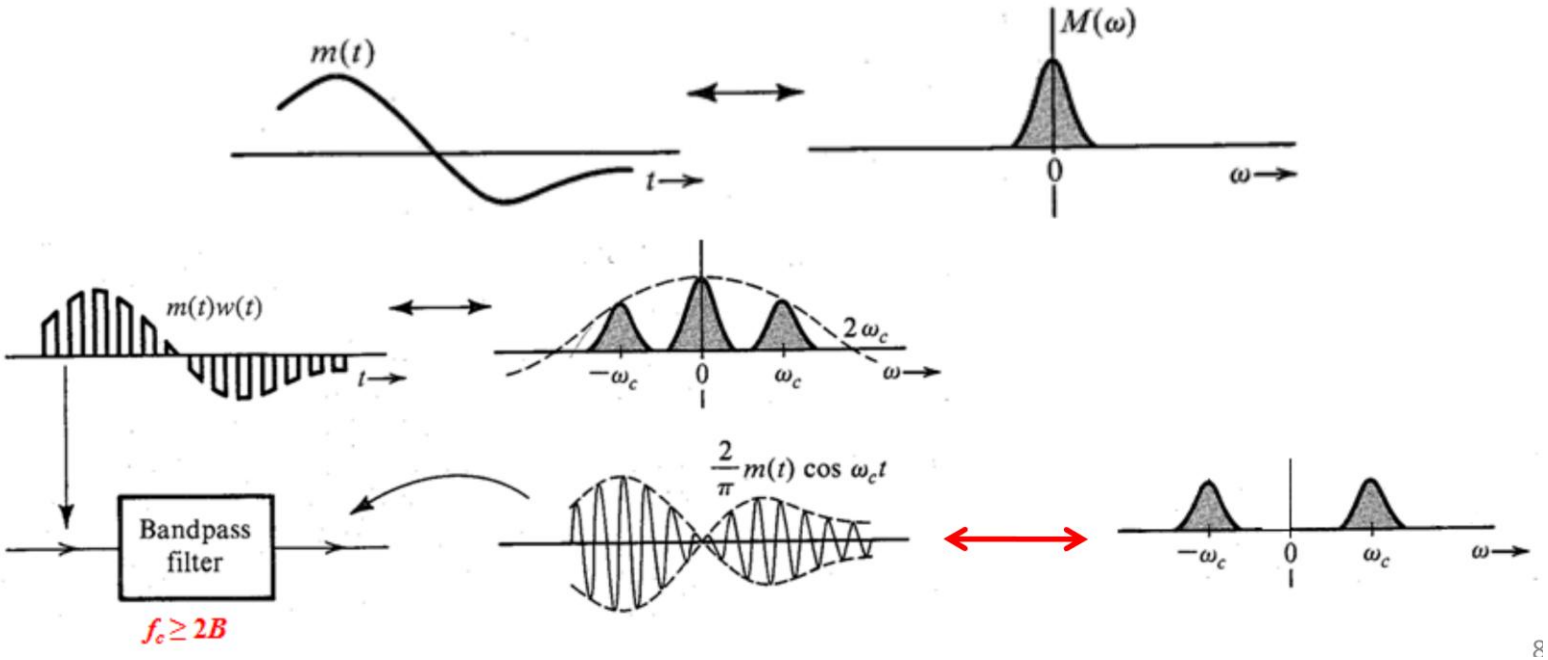
Passing this product through a **bandpass filter of bandwidth 2B Hz tuned to ω_c** gives the desired modulated signal $c_1\, m(t) \cos(\omega_c t + \theta_1)$.

DSB-SC Modulators (3): Square Wave w(t) as ϕ(t)

$$w(t) = \frac{1}{2} + \frac{2}{\pi}\left(\cos\omega_c t - \frac{1}{3}\cos 3\omega_c t + \frac{1}{5}\cos 5\omega_c t - \cdots\right)$$

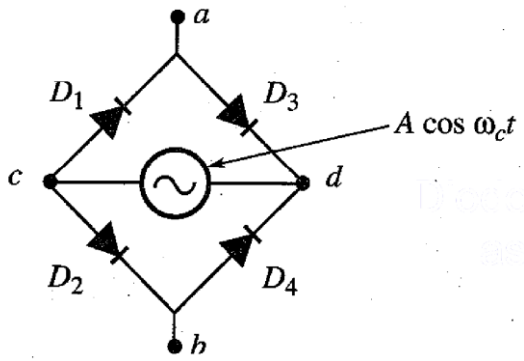


$$m(t)\,w(t) = \frac{1}{2}m(t) + \frac{2}{\pi}\left[m(t)\cos\omega_c t - \frac{1}{3}m(t)\cos 3\omega_c t + \frac{1}{5}m(t)\cos 5\omega_c t - \cdots\right]$$

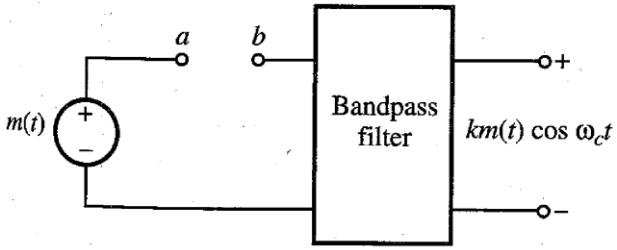


DSB-SC Modulators (4): Diode-Bridge Modulator

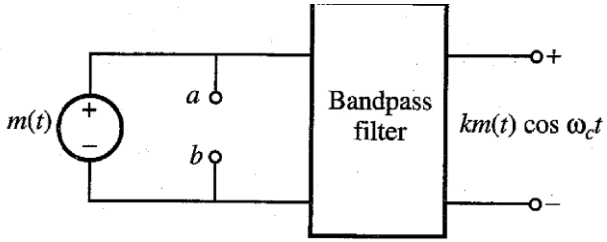
Switching modulator circuits: A. Diode-bridge modulator



The diode bridge works as a **switch** driven by the carrier.



Series-bridge diode modulator

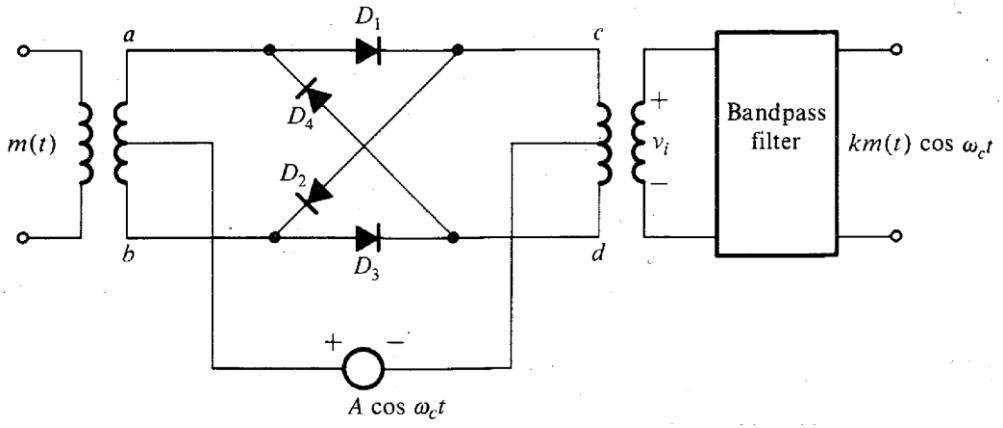


Shunt-bridge diode modulator

- When terminal **c** is positive with respect to **d**, all four diodes conduct.
- As D_1 and D_2 (also D_3 and D_4) are matched, terminals **a** and **b** are short-circuited.
- When terminal **d** is positive with respect to **c**, all four diodes are open.
- Thus **m(t) is switched at a frequency f_c**.
- It is a **single balanced modulator**.

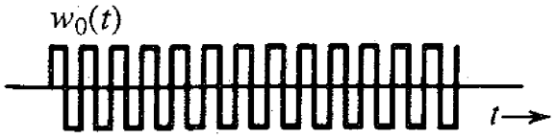
DSB-SC Modulators (5): Ring Modulator

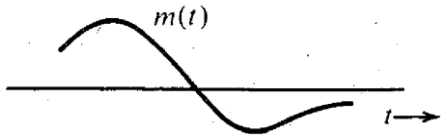
B. Ring modulator

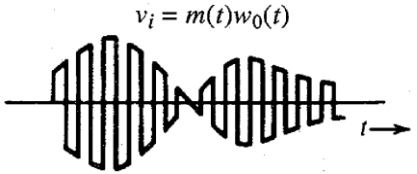


- **Positive half-cycle**: D_1 and D_3 conduct
- **Negative half-cycle**: D_2 and D_4 conduct
- m(t) is multiplied by a **bipolar** square wave $w_0(t) = 2w(t) - 1$.

A **double-balanced** modulator: both inputs are absent at the BPF input.



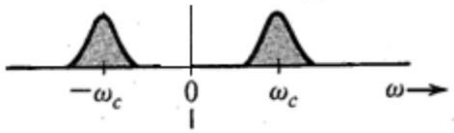




$$w_0(t) = 2w(t) - 1 = \frac{4}{\pi}\left(\cos\omega_c t - \frac{1}{3}\cos 3\omega_c t + \frac{1}{5}\cos 5\omega_c t - \cdots\right)$$

$$v_i(t) = m(t)\,w_0(t) = \frac{4}{\pi}\left[m(t)\cos\omega_c t - \frac{1}{3}m(t)\cos 3\omega_c t + \frac{1}{5}m(t)\cos 5\omega_c t - \cdots\right]$$

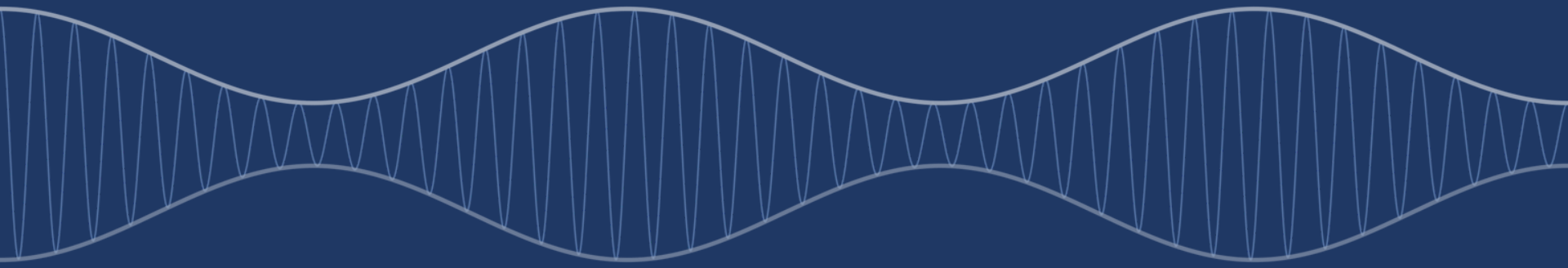
After the BPF: $\dfrac{4}{\pi}m(t)\cos\omega_c t$



PART 4

Bandwidth-Efficient AM

Single sideband (SSB-SC), quadrature amplitude modulation (QAM) and vestigial sideband (VSB)



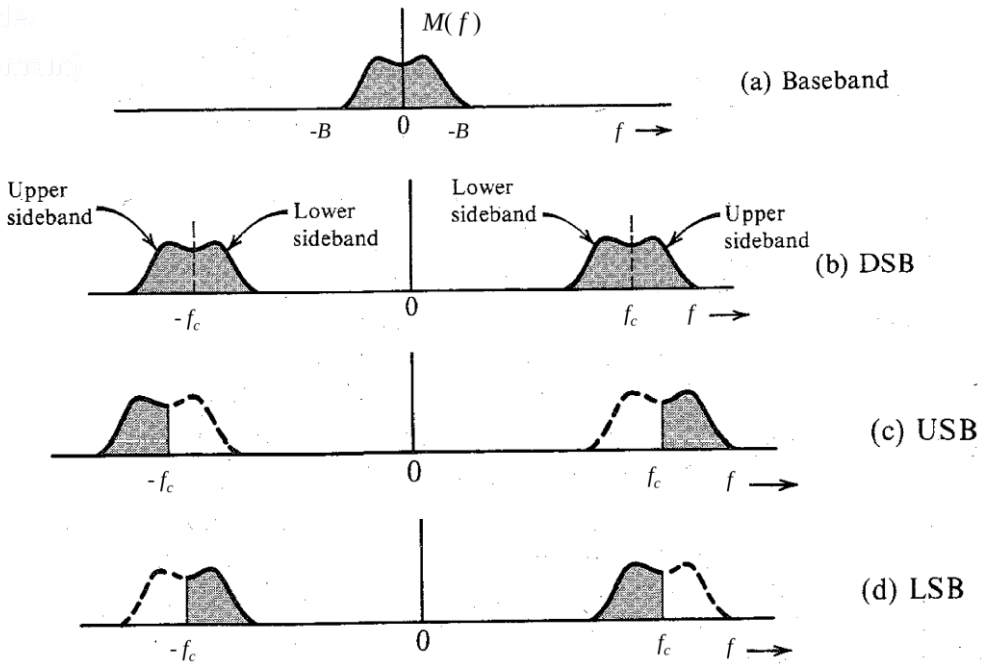
SSB-SC: Principle (1)

Single sideband, suppressed carrier

- Either the **USB or the LSB** is transmitted.
- **Required bandwidth: B Hz**.
- Spectral efficiency is improved by **100%** compared with DSB.

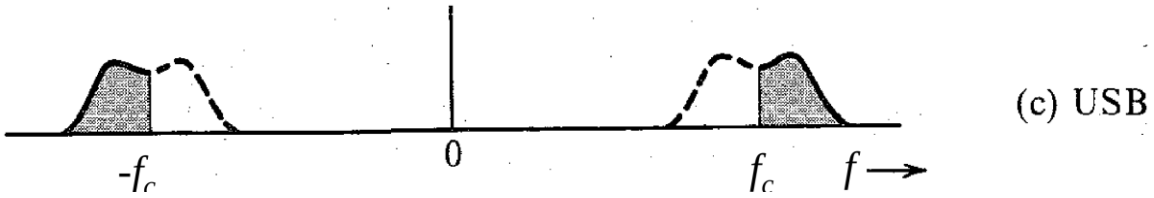
The two sidebands of DSB carry the **same information**, so transmitting one of them is enough.

Transmitter side (frequency domain)

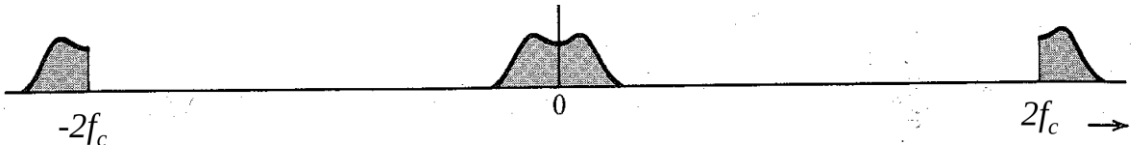


SSB-SC: Principle (2)

Receiver side (frequency domain): SSB can be **detected coherently**.



(c) Received USB signal



After multiplication by the carrier: a low-pass filter recovers the baseband message

Hilbert transformer

An **ideal phase shifter** that shifts the phase of **every spectral component by –π/2**. $m_h(t)$ is the Hilbert transform of m(t).

$$\varphi_{USB}(t) = m(t)\cos\omega_c t - m_h(t)\sin\omega_c t$$

$$\varphi_{LSB}(t) = m(t)\cos\omega_c t + m_h(t)\sin\omega_c t$$

$$\varphi_{SSB}(t) = m(t)\cos\omega_c t \mp m_h(t)\sin\omega_c t$$

Upper sign: USB; lower sign: LSB.

Comparison: AM, DSB-SC and SSB-SC

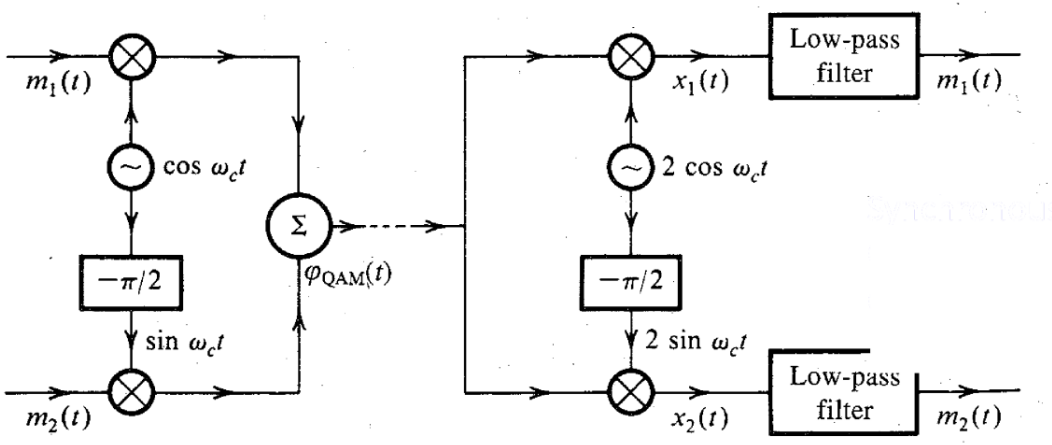
	AM (DSB-WC)	DSB-SC	SSB-SC
Modulation	Simple	Simple	Costly and complex (difficult to generate)
Demodulation	Envelope detection or coherent detection	Coherent detection: costly and complex	Coherent detection: costly and complex (envelope detection possible for SSB-WC, but not energy efficient)
Power efficiency	**Max 33%: inefficient**	**100%: efficient**	**100%: efficient**
Bandwidth requirement	**Twice the signal BW: inefficient**	**Twice the signal BW: inefficient**	**Equal to the signal BW: efficient**

Each step saves a resource (power, then bandwidth) at the cost of **more complex** transmitters and receivers.

QAM: Principle

- SSB signals are difficult to generate; **QAM is an attractive alternative** to SSB.
- Two baseband signals, each of bandwidth B Hz, are sent over the **same band of 2B Hz** (modulation: DSB-SC).
- The two carriers have the **same frequency** with a **phase difference of π/2**.
- QAM is also known as **quadrature multiplexing (QM)**.

$$\varphi_{QAM}(t) = m_1(t)\cos\omega_c t + m_2(t)\sin\omega_c t$$



Transmitter (left) and synchronous / coherent detector (right)

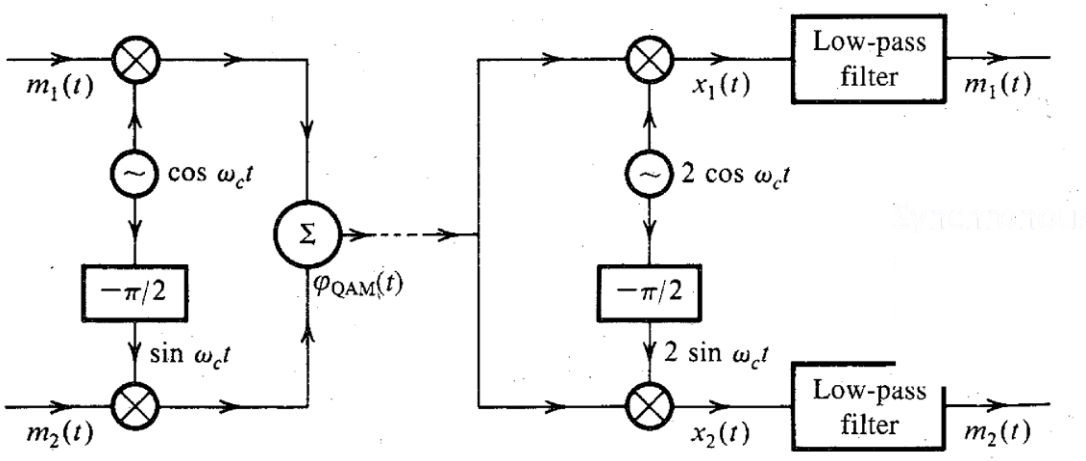
QAM: Detection (1)

In-phase (I) channel

$$x_1(t) = 2\varphi_{QAM}(t)\cos\omega_c t = m_1(t) + m_1(t)\cos 2\omega_c t + m_2(t)\sin 2\omega_c t$$

Quadrature (Q) channel

$$x_2(t) = 2\varphi_{QAM}(t)\sin\omega_c t = m_2(t) - m_2(t)\cos 2\omega_c t + m_1(t)\sin 2\omega_c t$$



- The low-pass filters remove the terms at **$2\omega_c$**.
- Outputs: **$m_1(t)$** from the I channel and **$m_2(t)$** from the Q channel.
- Requires a **synchronous** local carrier.

QAM: Detection (2): Loss of Synchronization

Impact of a local-carrier phase error θ

Loss of power in the wanted signal (the factor cos θ) and **interference** (crosstalk) from the other channel (the factor sin θ).

I-channel multiplier output with phase error θ

$$x_1(t) = 2[m_1(t)\cos\omega_c t + m_2(t)\sin\omega_c t]\cos(\omega_c t + \theta)$$

$$= m_1(t)\cos\theta + m_1(t)\cos(2\omega_c t + \theta) - m_2(t)\sin\theta + m_2(t)\sin(2\omega_c t + \theta)$$

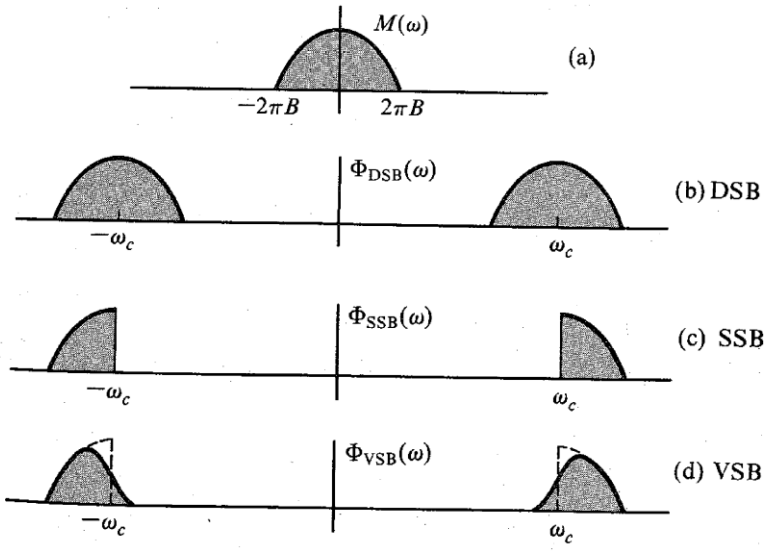
Output of the I channel (after the LPF):

$$m_1(t)\cos\theta - m_2(t)\sin\theta$$

Exercise: derive the output of the quadrature (Q) channel with the same phase error θ.

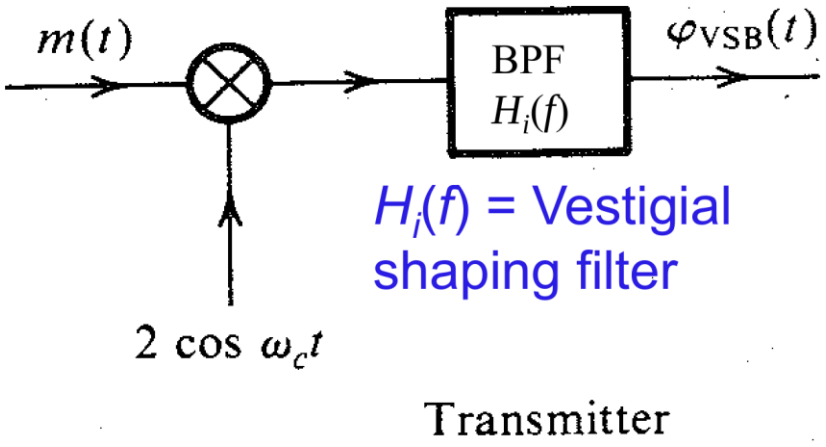
VSB (1): Principle

- SSB signals are **difficult to generate**, and DSB requires **twice the signal bandwidth**.
- VSB is a **compromise** between DSB and SSB.
- VSB inherits the advantages of DSB and SSB, but avoids their disadvantages at a small cost.
- The bandwidth of VSB is a little (typically **25%**) greater than SSB.
- VSB is also known as an **asymmetric sideband system**.

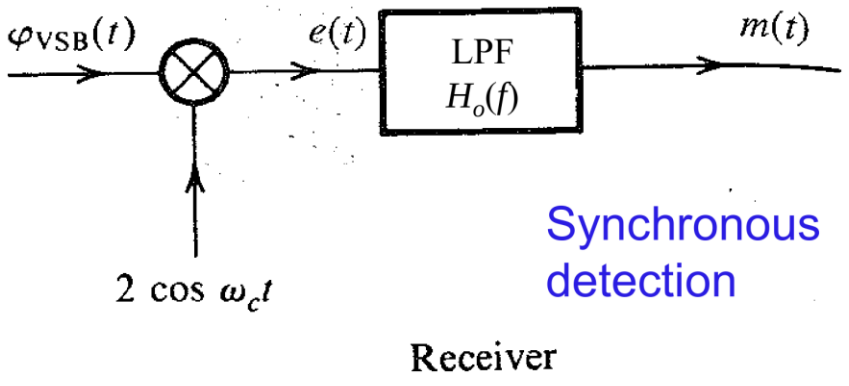


"Vestigial" means remaining, leftover or residual.

VSB (2): Generation and Detection



$H_i(f)$ = Vestigial shaping filter



Synchronous detection

VSB signal generation

$$\Phi_{VSB}(f) = [M(f-f_c) + M(f+f_c)]H_i(f)$$

Coherent detection

$$e(t) = 2\varphi_{VSB}(t)\cos\omega_c t \iff \Phi_{VSB}(f+f_c) + \Phi_{VSB}(f-f_c)$$

$$M(f) = [\Phi_{VSB}(f+f_c) + \Phi_{VSB}(f-f_c)]H_o(f)$$

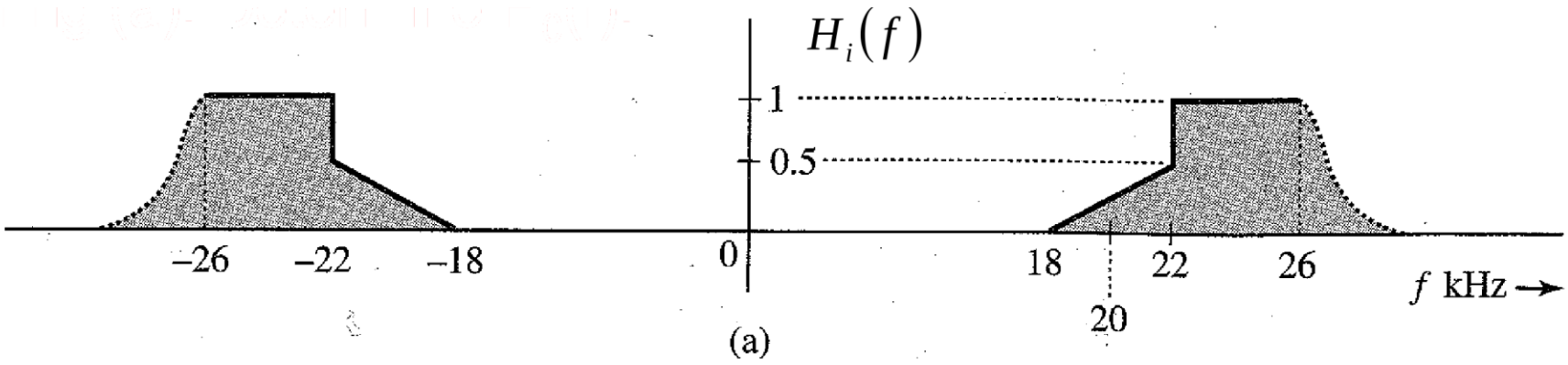
After suppressing the signal at $2\omega_c$ with the LPF $H_o(f)$:

$$H_o(f) = \frac{1}{H_i(f+f_c) + H_i(f-f_c)}, \qquad |f| \le B$$

VSB (3): Example

PROBLEM

The carrier frequency is 20 kHz and the baseband signal bandwidth is 6 kHz. The vestigial shaping filter $H_i(f)$ is shown in Fig. (a). Determine the output filter $H_o(f)$ for distortionless reception.



Carrier frequency	Message bandwidth	$H_i(f)$ from 18 to 22 kHz	$H_i(f)$ from 22 to 26 kHz
f_c = 20 kHz	**B = 6 kHz**	**rises linearly 0 → 0.5**	**= 1**

VSB (3): Solution

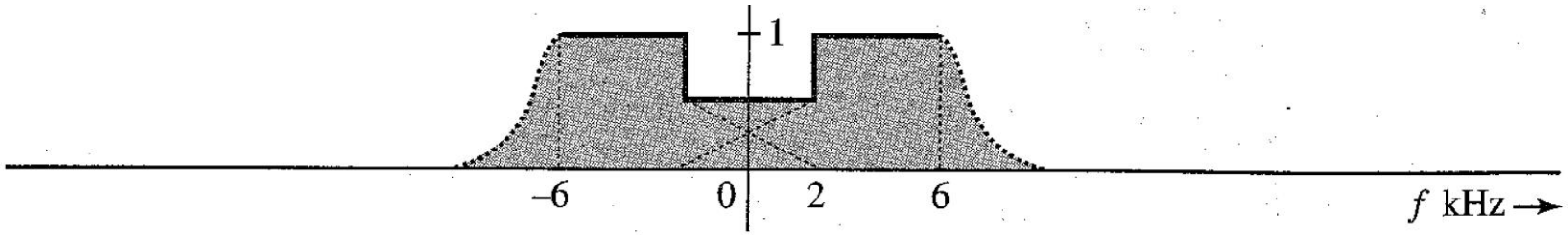
1 **Output filter required for distortionless detection**

$H_o(f) = 1 / [H_i(f + f_c) + H_i(f - f_c)]$ for $|f| \le B = 6$ kHz

2 **Add the two shifted copies of $H_i(f)$ ($f_c = 20$ kHz)**

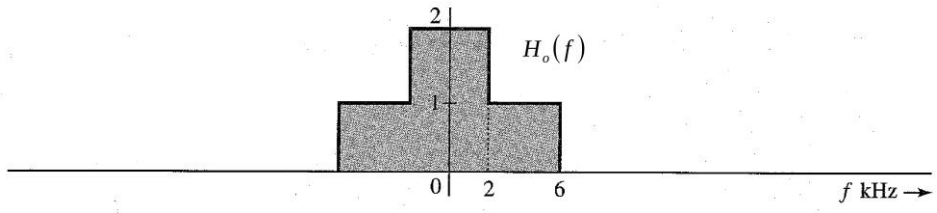
$2 < |f| \le 6$ kHz: one copy is 1 (22 to 26 kHz), the other is 0 $\Rightarrow$ sum = **1**

$|f| < 2$ kHz: the two ramps (0 → 0.5 and 0.5 → 0) add to a constant $\Rightarrow$ sum = **0.5**



3 **Invert the sum**

$H_o(f) =$ **2** for $|f| < 2$ kHz, $H_o(f) =$ **1** for $2 \le |f| \le 6$ kHz



$H_o(f)$ boosts the vestige region by 2 so every message frequency is recovered with the same gain.

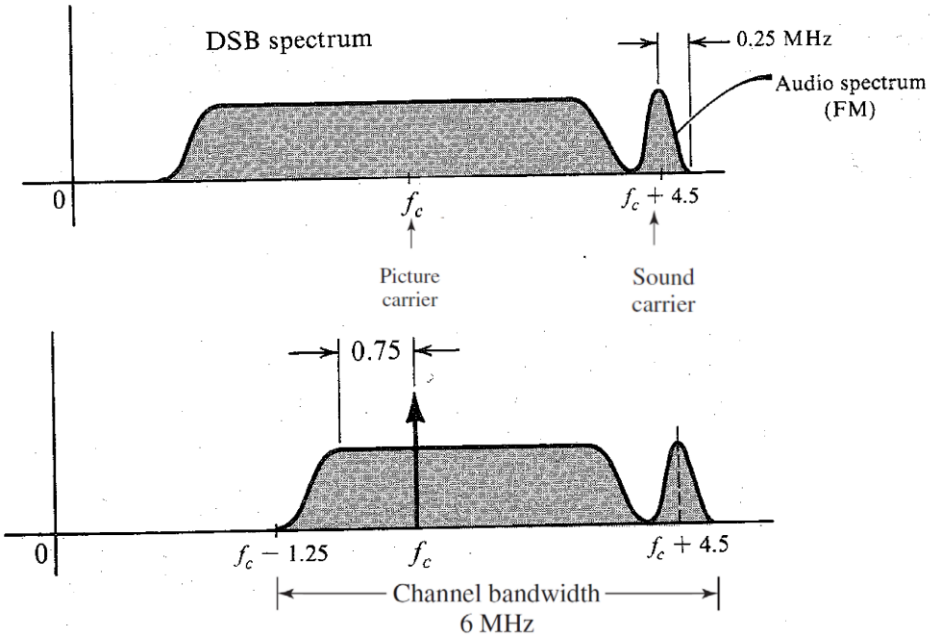
VSB Application: Analog Television

Video signal:

- large bandwidth (4.5 MHz): DSB would require **9 MHz**
- contains a significant low-frequency component: **SSB is not feasible**

The demodulation of the TV signal must be simple and cost effective, so an **envelope detector** is preferred.

So, **VSB modulation with the carrier** is chosen for TV broadcast.



Top: DSB spectrum. Bottom: transmitted VSB spectrum (6 MHz channel)

VSB Application: Digital Television

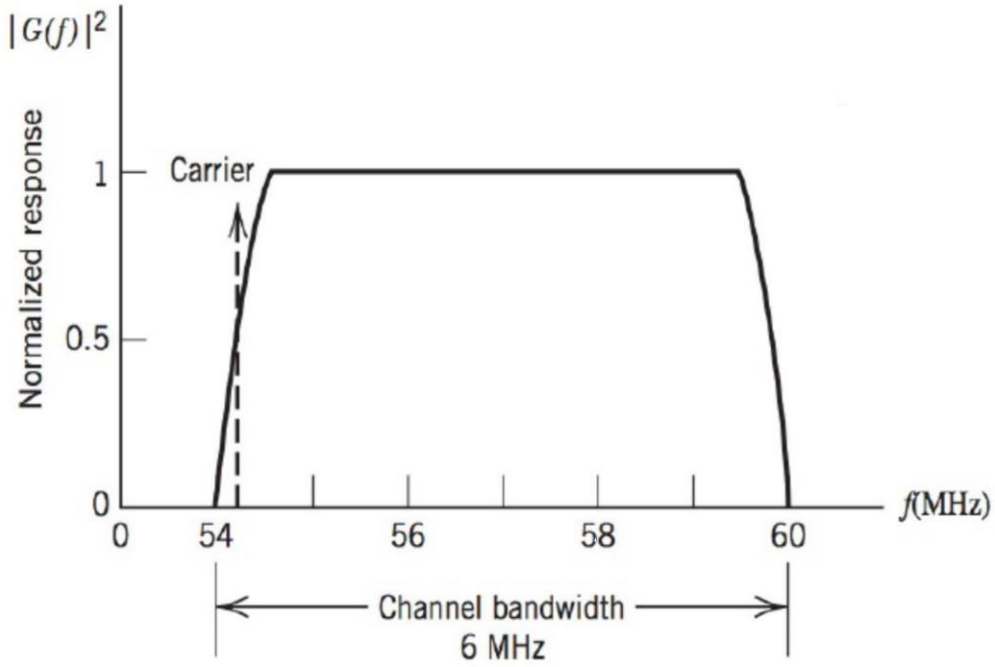


FIGURE 3.21 Idealized amplitude spectrum of VSB modulated digital television signal.

One data stream

With the digital approach, the **sound, video and colour** are all integrated as one data stream.

Shaped spectrum

The spectrum extends **0.31 MHz below** the carrier and **5.69 MHz above** it: 6 MHz in total.

Key Takeaways

1. **Modulation** moves the message to the channel's band: it enables multiplexing (FDM, TDM) and practical antenna sizes (about 3 m instead of hundreds of km for speech on a 10 MHz carrier).

2. **AM (DSB-WC)**: $\phi_{AM}(t) = [A_c + m(t)] \cos \omega_c t$, bandwidth 2B, envelope detection needs $0 \leq \mu \leq 1$, and power efficiency is at most 33%.

3. **DSB-SC** removes the carrier for 100% power efficiency, but needs coherent detection. Switching and ring modulators are practical generators.

4. **SSB-SC** halves the bandwidth to B; **QAM** sends two messages in 2B; **VSB** is the DSB/SSB compromise used for TV.

5. **VSB detection** needs $H_o(f) = 1/[H_i(f + f_c) + H_i(f - f_c)]$ over the message band.